Patient educator training

Supporting families with MPS IVA and MPS VI: from diagnosis to adult care

Focus on MPS IVA (Morquio A) and MPS VI (Maroteaux-Lamy) for staff following patients in the support program

Medical Affairs | October 2026

For internal training of patient support program staff only. Not for distribution to patients. Product information may vary by country; always refer to the local prescribing information.

01 Your role

02 Diagnosis

03 Understanding MPS

04 Starting treatment

05 Infusion day

06 Follow-up

07 Living with MPS

08 Pharmacovigilance

Agenda

01 Your role
Boundaries, compliance and each family contact

02 Diagnosis
First signs, how it is confirmed, family testing

03 Understanding MPS
Inheritance, organs affected, MPS IVA and VI

04 Starting treatment
Care options, baseline tests, what ERT achieves

05 Infusion day
Each infusion, side effects, managing reactions

06 Ongoing follow-up
Tests, MDT visits, adherence, warning signs

07 Living with MPS
Surgery safety, daily life, move to adult care

08 Pharmacovigilance
Reporting AEs, special situations, complaints

Learning objectives

- Explain MPS IVA and MPS VI and their inheritance in plain words
- Stay within the program's role, compliance and privacy rules
- Guide families through each step, from diagnosis to adult care
- Recognise warning signs and infusion reactions, and escalate the same day
- Support adherence, follow-up tests and MDT visits at every contact
- Report adverse events, special situations and complaints on time

AE, adverse event; ERT, enzyme replacement therapy; MDT, multidisciplinary team; MPS, mucopolysaccharidosis

01

Your role

What the program asks of you, its limits, and how to run every family contact

In this section

Your role and boundaries

Each family contact, step by step

Your role: what you do, and what you never do

You do

- Educate families using program-approved materials only
- Remind: infusion dates, tests and MDT visits
- Listen, and record every contact accurately
- Escalate warning signs to the treating team the same day
- Report every AE, special situation and complaint

You never

- Give medical advice or change doses, rates or premedication
- Promote products or discuss off-label use
- Promise outcomes or question the physician's plan
- Share patient data outside the program
- Accept gifts or payments from families

Before the first contact

- Signed consent from the patient or caregiver
- Referral or agreement from the treating physician
- Data privacy rules applied to every record
- PV training completed, and refreshed yearly
- Escalation and PV contacts saved

The program is educational and non-promotional. If in doubt, stop and ask your program manager or the treating physician.

AE, adverse event; MDT, multidisciplinary team; PV, pharmacovigilance

EMA. Guideline on good pharmacovigilance practices (GVP) Module VI; ABPI. Guidance notes for patient safety and pharmacovigilance in patient support programmes; local industry code of practice.

Each family contact, step by step

1 Prepare

- Read the notes from the last contact
- Check the last infusion and upcoming tests
- Confirm language and who will join

2 Open

- Ask how the patient and family are doing
- Explain why you are calling and how long it takes

3 Explore

- Ask open questions and listen
- Explore barriers without blame (motivational interviewing)

4 Educate

- One or two key messages per contact
- Plain words, no jargon
- Check understanding with teach-back

5 Close

- Agree next steps and dates
- Document the contact
- Escalate or report if needed

Teach-back: ask the family to explain it in their own words. If they can't, the message wasn't clear, so try again another way.

Respect culture and family roles, involve the patient more as they grow, and respond to anger or denial with patience and support.

Akyol MU et al. Orphanet J Rare Dis 2019;14:137; Akyol MU et al. Orphanet J Rare Dis 2019;14:118; Lampe C et al. Mol Genet Metab Rep 2019;21:100508.

02

First signs and diagnosis

Why diagnosis takes years, how it is confirmed and what it means for the family

In this section

Time to diagnosis

How MPS IVA and MPS VI are diagnosed

Time to diagnosis

~4 years

average delay from first symptoms to diagnosis of MPS[1]

Early signs are often subtle or mistaken for common childhood conditions.

1 Earlier treatment, better outcomes

In MPS VI siblings, the child who started ERT at 7.5 weeks had no scoliosis at age 3.6 years; the older sibling, untreated until then, already had scoliosis.[2]

2 Guidelines: start ERT at diagnosis

International recommendations for MPS IVA and MPS VI advise starting ERT as soon as the diagnosis is confirmed.[3,4]

3 Established damage does not reverse

ERT slows or stabilises disease, but existing bone and joint changes remain, so every delay costs function.[3,4]

ERT, enzyme replacement therapy; MPS, mucopolysaccharidosis

1. Mitchell J et al. Presenting signs and symptoms of MPS: systematic literature analysis. WORLDSymposium 2018, poster 244. 2. McGill JJ et al. Clin Genet 2010;77:492–8. 3. Akyol MU et al. Orphanet J Rare Dis 2019;14:137. 4. Akyol MU et al. Orphanet J Rare Dis 2019;14:118.

How MPS IVA and MPS VI are diagnosed

1 Suspected

- **Signs:** short stature, skeletal or joint changes, corneal clouding, repeated ear or chest infections
- **Urine GAG test:** can raise suspicion but cannot confirm MPS

2 Confirmed

- **Enzyme test** in blood cells: the gold standard
- **Dried blood spot:** easy to send, but needs confirmation
- **Genetic test:** confirms the variant and the MPS type

3 Family follow-up

- **Siblings** should be offered testing; early treatment matters
- **Genetic counselling:** carrier, premarital and prenatal options
- **Newborn screening** is not routine in most countries

Your role: explain the steps in plain words and refer to the treating physician and genetic counsellor. You never order or interpret tests.

GAG, glycosaminoglycan; MPS, mucopolysaccharidosis. Laboratory detail is in the appendix

Wood T et al. Mol Genet Metab 2012;106:73–82; Akyol MU et al. Orphanet J Rare Dis 2019;14:137; Akyol MU et al. Orphanet J Rare Dis 2019;14:118.

03

Understanding MPS

What families need to understand after the diagnosis

In this section

What MPS is

How MPS is inherited

Organ systems affected

MPS IVA (Morquio A)

MPS VI (Maroteaux-Lamy)

What is MPS?



Both copies of an MPS gene carry a variant (one copy in males with X-linked MPS II)

The lysosomal enzyme that breaks down GAGs is missing or works poorly

GAGs (long sugar chains) accumulate inside lysosomes

Storage triggers inflammation and disrupts cell and tissue function

Bones, joints, airway, heart, eyes and ears are progressively affected

~1 in 25,000[1]

live births are affected by an MPS disorder, with wide regional variation

From birth[2]

the enzyme defect is present at birth; most signs appear over months to years

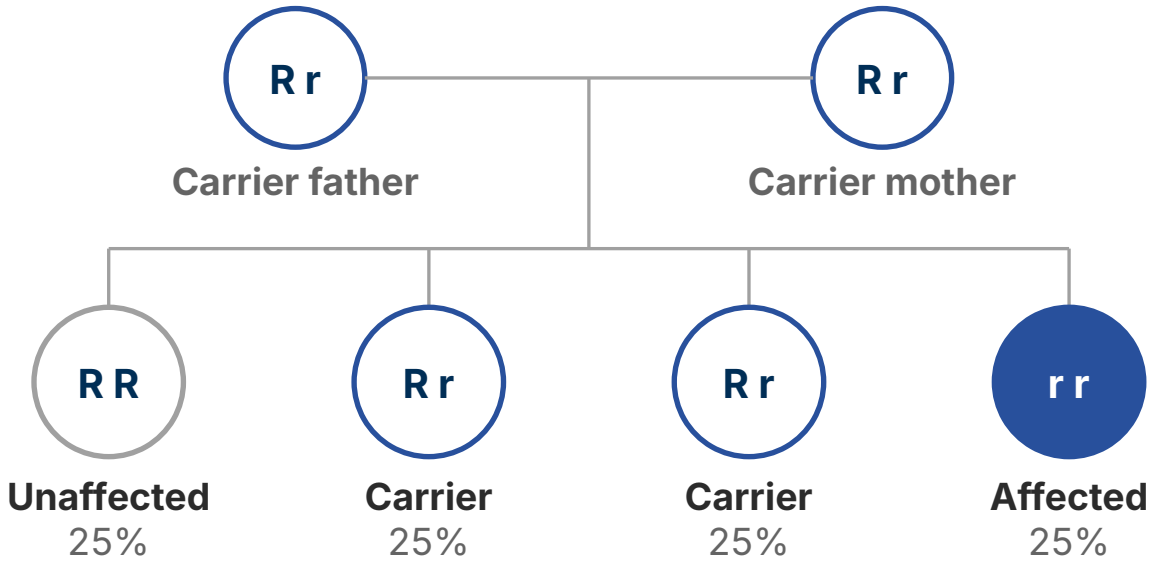
Lifelong[3]

storage continues without treatment, so the disease progresses with age

GAG, glycosaminoglycan; MPS, mucopolysaccharidosis

1. Gökdoğan C et al. Braz J Otorhinolaryngol 2016;82:281–4. 2. Muenzer J. Rheumatology (Oxford) 2011;50(Suppl 5):v4–12. 3. Neufeld EF, Muenzer J. In: The Metabolic and Molecular Bases of Inherited Disease. 8th ed. McGraw-Hill; 2001:3421–52.

Inheritance of MPS



R working gene copy **r** copy with the disease variant

The same odds apply to every pregnancy, whatever the outcome of earlier pregnancies.

MPS II is the exception

X-linked: a carrier mother has a 50% chance of passing the variant to each son (affected) and to each daughter (usually a carrier).

What this means for families

- **Consanguinity** raises the chance that both parents carry the same variant
- **Siblings** of a patient should be offered testing, as early treatment matters
- **Genetic counselling** supports family planning, including premarital and prenatal options

MPS, mucopolysaccharidosis

Muenzer J. Rheumatology (Oxford) 2011;50(Suppl 5):v4–12; Akyol MU et al. Orphanet J Rare Dis 2019;14:137; Akyol MU et al. Orphanet J Rare Dis 2019;14:118.

Organ systems affected in MPS

Bones and joints

Dysostosis multiplex, short stature, stiff (or lax in IVA) joints, hip and knee deformity

Spine and nerves

Cord compression, carpal tunnel; brain affected in MPS I, II, III, VII

Airway and lungs

Narrow airway, sleep apnoea, small rigid chest, recurrent infections

Heart

Thickened valves that leak or narrow; cardiomyopathy

Eyes

Corneal clouding, glaucoma, retinal and optic-nerve disease

Ears

Recurrent ear infections; conductive and sensorineural hearing loss

Abdomen

Enlarged liver and spleen; umbilical and inguinal hernias

Teeth and mouth

Widely spaced teeth, enamel defects, enlarged gums and tongue

GAG, glycosaminoglycan; MPS, mucopolysaccharidosis

Neufeld EF, Muenzer J. In: The Metabolic and Molecular Bases of Inherited Disease. 8th ed. McGraw-Hill; 2001:3421–52; Muenzer J. Rheumatology (Oxford) 2011;50(Suppl 5):v4–12; Clarke L et al. J Inborn Errors Metab Screen 2018;6:1–12.

MPS IVA (Morquio A): key facts

Key facts

Enzyme
GALNS is deficient

Stored GAGs
Keratan sulfate and chondroitin-6-sulfate

Onset
Normal at birth; skeletal signs develop progressively from about age 3

Cognition
Usually normal

Life expectancy
From the second decade to near-normal, depending on severity

Image: patient with MPS IVA. Not all signs or symptoms are present in every patient.

Ophthalmological
Diffuse corneal clouding, cataracts, astigmatism, visual impairment[1,2]

Pulmonary
Obstructive sleep apnoea, respiratory infections, respiratory failure[2,3]

Gastrointestinal
Hepatomegaly, splenomegaly, hernias, diarrhoea, constipation[1,2]

Neurological
Cervical myelopathy from odontoid dysplasia or cervical spine instability, tetraplegia[1,3]



ENT
Conductive and neurosensory hearing loss, airway obstruction[1,2]

Dental
Widely spaced teeth, thin enamel, frequent caries, pointed cusps, spade-shaped incisors, abnormal buccal surfaces[2,3]

Cardiac
Mitral and aortic valve stenosis and regurgitation, left ventricular hypertrophy[2]

Skeletal
Short stature, spinal abnormalities, atlantoaxial instability, hip and odontoid dysplasia, genu valgum, pectus carinatum, joint abnormalities[1]

ENT, ear, nose and throat; GAG, glycosaminoglycan; GALNS, N-acetylgalactosamine-6-sulfatase; MPS, mucopolysaccharidosis

Image: 1. Hendriksz CJ et al. Am J Med Genet A 2015;167A:11–25. 2. Hendriksz CJ et al. J Inherit Metab Dis 2013;36:309–22. 3. Tomatsu S et al. Curr Pharm Biotechnol 2011;12:931–45. Key facts: Hendriksz CJ et al. Am J Med Genet A 2015;167A:11–25; Hendriksz CJ et al. Mol Genet Metab 2013;110:54–64; Akyol MU et al. Orphanet J Rare Dis 2019;14:137.

MPS IVA progression: two patients from infancy to adulthood



MPS, mucopolysaccharidosis

1. Hendriksz CJ et al. Am J Med Genet A 2015;167A:11–25.

MPS IVA: morbidity and mortality

Morbidity[1,2]

- Progressive multisystem manifestations[1]
  - Cardiac disease
  - Respiratory disease
  - Musculoskeletal disease
  - Spinal cord compression
- Multiple surgeries[2]
- Reduced mobility[1]
- Chronic pain[2]
- Lower quality of life[2]

Mortality[2,3]

Life expectancy ranges from the second decade to a near-normal lifespan

Causes of death include:

- Pulmonary complications
- Neurological complications
- Cardiac complications

MPS, mucopolysaccharidosis

1. Hendriksz CJ et al. Mol Genet Metab 2013;110:54–64. 2. Harmatz P et al. Mol Genet Metab 2013;109:54–61. 3. Tomatsu S et al. Curr Pharm Biotechnol 2011;12:931–45.

MPS VI (Maroteaux-Lamy): key facts

Key facts

Enzyme
ARSB is deficient

Stored GAG
Dermatan sulfate

Spectrum
Rapid forms show signs before age 2; slower forms may be diagnosed in adulthood

Genetics
>130 ARSB variants; symptoms appear below ~10% of normal enzyme activity

Cognition
Usually normal

Image: patient with MPS VI. Not all signs or symptoms are present in every patient.

ENT
Impaired hearing, recurrent otitis media, recurrent sinusitis[1–3]

Neurological
Spinal cord compression, hydrocephalus, carpal tunnel syndrome[1]

Pulmonary
Obstructive and restrictive airway disease, sleep apnoea, recurrent pulmonary infections[2,3]

Abdominal
Hepatosplenomegaly, umbilical and inguinal hernias[4,5]



Ophthalmological
Impaired vision, corneal clouding, glaucoma, optic nerve disease[1,5]

Dental
Enlarged tongue, small and widely spaced teeth[6]

Cardiac
Valve disease, cardiomyopathy, cardiac arrhythmia, systemic and pulmonary hypertension[1,7]

Skeletal
Joint stiffness and contractures, dysostosis multiplex, hip dysplasia[1]

ARSB, arylsulfatase B (N-acetylgalactosamine 4-sulfatase); ENT, ear, nose and throat; GAG, glycosaminoglycan; MPS, mucopolysaccharidosis

Giugliani R et al. Pediatrics 2007;120:405–18; D'Avanzo F et al. Int J Mol Sci 2021;22:13456; Akyol MU et al. Orphanet J Rare Dis 2019;14:118.

MPS VI: classical and non-classical disease





Feature	Classical	Non-classical
Enzyme activity[1]	No significant activity	Some residual activity
uGAG[2]	>200 µg/mg	<100 µg/mg
Onset of symptoms[3]	<3 years	>5 years
Height[1]	<95–100 cm	>140 cm
Debilitation[1]	First decade of life	Second decade or later
Death[2,3]	<20 years	Survival into adulthood
Share of patients[3]	60%	40%

Debilitation, weakness. GAG, glycosaminoglycan; MPS, mucopolysaccharidosis; uGAG, urinary GAG (µg/mg creatinine)

1. Giugliani R et al. Pediatrics 2007;120:405–18. 2. Swiedler SJ et al. Am J Med Genet A 2005;134A:144–50. 3. Montaño AM et al. J Inherit Metab Dis 2007;30:165–74.

MPS VI: rapidly and slowly progressing examples

Rapidly progressing (classical)

Features are apparent from an early age

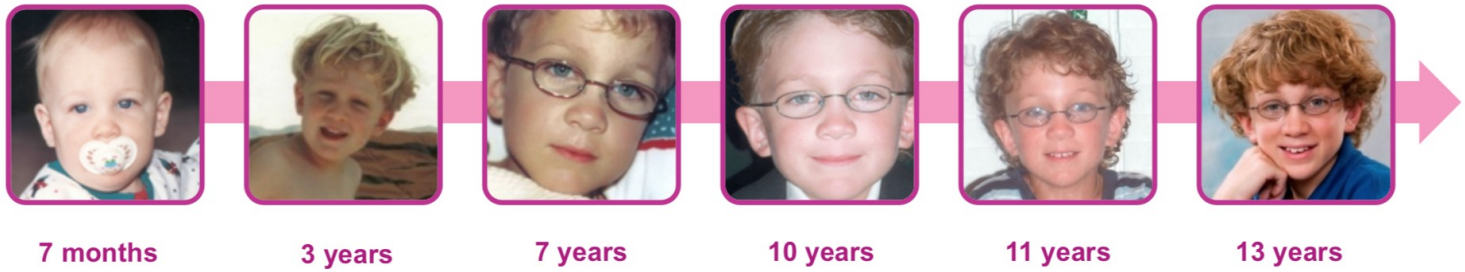
Patient followed from birth to 26 years



Slowly progressing (non-classical)

No obvious clinical features through early adolescence

Patient followed from 7 months to 13 years



MPS, mucopolysaccharidosis

Giugliani R et al. Pediatrics 2007;120:405–18.

MPS VI: morbidity and mortality

Morbidity[1]

- Physical and functional well-being decline, limiting endurance and mobility
  - Skeletal deformities
  - Joint disease
  - Cardiopulmonary disease
  - Blindness
  - Spinal cord compression

Mortality[1]

- **Severely affected patients:** live to their early teens or early 20s
- **Slowly progressing patients:** can live into their 40s or 50s

Causes of death include:

- Infections
- Surgical complications
- Cardiopulmonary disease

MPS, mucopolysaccharidosis

1. Giugliani R et al. Pediatrics 2007;120:405–18.

04

Starting treatment

Care options, baseline tests and what ERT can and cannot do

In this section

Management of MPS

Baseline assessment before ERT

Enzyme replacement therapy

What ERT can achieve

Management of MPS

Disease-specific therapy

- **Enzyme replacement therapy (ERT)** is first-line for MPS IVA and MPS VI: a weekly, lifelong intravenous infusion
- **Haematopoietic stem cell transplant** is standard for severe MPS I; it is investigational in MPS IVA and MPS VI

Supportive and surgical care

- **Surgery:** cervical decompression and fusion, hip and knee, ENT, heart valve, cornea
- **Breathing support:** CPAP or BiPAP for sleep apnoea, oxygen when needed
- **Daily function:** pain control, physiotherapy, hearing aids, dental care, psychosocial support

Under investigation

- **Gene therapy** and other new approaches are in clinical research
- **Not approved** for MPS IVA or MPS VI; discuss only if the treating physician raises it

ERT works alongside supportive care, not instead of it: patients on ERT still need regular specialist review and often surgery.

BiPAP, bilevel positive airway pressure; CPAP, continuous positive airway pressure; ENT, ear, nose and throat; ERT, enzyme replacement therapy; MPS, mucopolysaccharidosis

Akyol MU et al. Orphanet J Rare Dis 2019;14:137; Akyol MU et al. Orphanet J Rare Dis 2019;14:118; Muenzer J. Rheumatology (Oxford) 2011;50(Suppl 5):v4–12.

Baseline assessment before ERT

Assessment	Why at baseline	MPS IVA (before elosulfase alfa)	MPS VI (before galsulfase)
Growth and general health	Reference for growth and daily function	Height, weight; health assessment questionnaire	Height, weight
Urinary biomarker	Starting level to judge ERT response	Urinary keratan sulfate (KS)	Urinary GAGs
Spine and brain imaging	Detect cord compression early	MRI of the whole spine (neutral position)	MRI of the brain and spine
Lung function	Starting breathing capacity	FVC and MVV	FVC and FEV1
Endurance	Starting walking capacity	6MWT; 3-minute stair climb	12MWT
Heart	Valve and heart muscle disease	ECG and echocardiography	ECG and echocardiography

Vision and hearing are also checked at baseline. The treating physician decides the final list for each patient.

6MWT/12MWT, 6-/12-minute walk test; ECG, electrocardiogram; ERT, enzyme replacement therapy; FEV1, forced expiratory volume in 1 s; FVC, forced vital capacity; GAG, glycosaminoglycan; KS, keratan sulfate; MRI, magnetic resonance imaging; MVV, maximum voluntary ventilation

Akyol MU et al. Orphanet J Rare Dis 2019;14:137; Akyol MU et al. Orphanet J Rare Dis 2019;14:118.

Enzyme replacement therapy (ERT)

	Elosulfase alfa	Galsulfase
Treats	MPS IVA (Morquio A)	MPS VI (Maroteaux-Lamy)
Replaces	GALNS	ARSB
Dose	2 mg/kg body weight, once weekly	1 mg/kg body weight, once weekly
Infusion time	About 4 hours (3.5–4.5 h, by volume)	At least 4 hours, slower in the first hour
Premedication	Antihistamine, with or without antipyretic, 30–60 minutes before the infusion	
Preparation	Dilute in 0.9% sodium chloride (100 or 250 mL by weight); rotate gently, never shake; infuse via a low-protein-binding 0.2 µm filter	
Storage	Unopened vials at 2–8 °C; do not freeze; use the diluted solution promptly	

Doses, rates and equipment always follow the local prescribing information and the treating centre's infusion protocol.

ARSB, arylsulfatase B; ERT, enzyme replacement therapy; GALNS, N-acetylgalactosamine-6-sulfatase; MPS, mucopolysaccharidosis

Elosulfase alfa Summary of Product Characteristics (EMA); Galsulfase Summary of Product Characteristics (EMA); Concolino D et al. Ital J Pediatr 2018;44(Suppl 2):120.

ERT efficacy versus placebo at 24 weeks

Elosulfase alfa | MPS IVA

+22.5 m

more distance walked in 6 minutes than with placebo at week 24 (phase 3, n=176)[1]

Galsulfase | MPS VI

+92 m

more distance walked in 12 minutes than with placebo at week 24 (phase 3, n=39); urinary GAGs fell by about 75%[2]

Long-term data

Registries following patients for up to 10 years (MPS IVA) and 15 years (MPS VI) report sustained falls in urinary GAGs and maintained endurance on continued weekly ERT.[3,4]

ERT, enzyme replacement therapy; GAG, glycosaminoglycan; MPS, mucopolysaccharidosis

1. Hendriksz CJ et al. J Inherit Metab Dis 2014;37:979–90. 2. Harmatz P et al. J Pediatr 2006;148:533–9. 3. Mitchell JJ et al. MARS 10-year data. WORLDSymposium 2025, poster. 4. Burton BK et al. Mol Genet Metab 2025;145:109135.

MPS IVA: long-term elosulfase alfa versus untreated patients

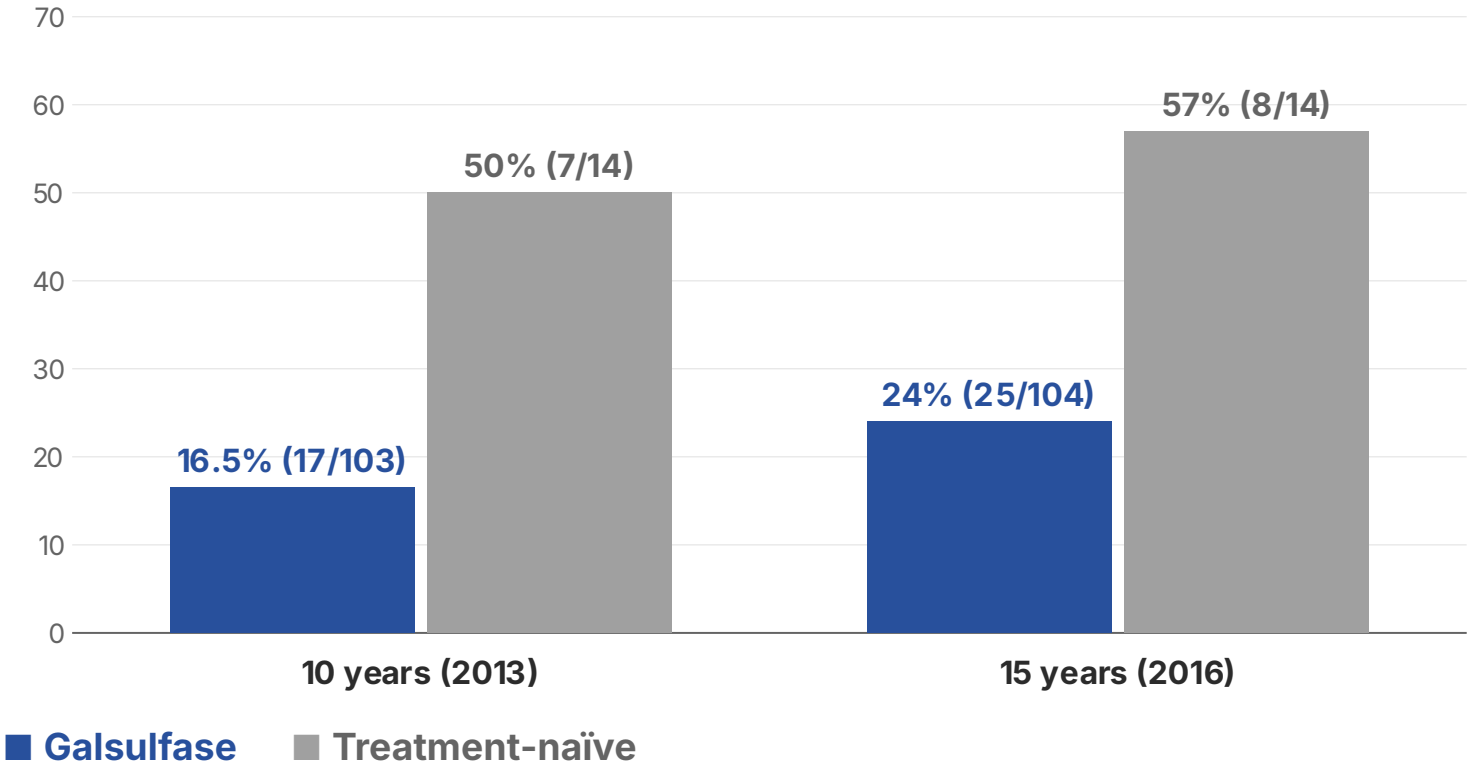
	Elosulfase alfa (MOR-004/005)	**Untreated** (MorCAP)
Endurance	Sustained improvement	Sustained deterioration
Urinary KS	Sustained reduction	Stable
Respiratory function	Sustained increase in patients aged ≤14 and >14 years	Increase in patients aged ≤14 years; deterioration in patients >14 years

KS, keratan sulfate; MorCAP, Morquio A Clinical Assessment Program

1. Hendriksz CJ et al. Mol Genet Metab 2016;119:131–43. 2. Hendriksz CJ et al. J Inherit Metab Dis 2016;39:839–47.

MPS VI: survival with galsulfase at 10 and 15 years

Mortality since the Survey Study, % of patients



Hazard ratio

HR 0.35

65% lower risk of death with galsulfase[2]

0.348 (95% CI 0.156–0.776), unadjusted

Kaplan-Meier survival

Curves separated in favour of galsulfase[2]

Log-rank P=0.0050; Wilcoxon P=0.0012

CI, confidence interval; HR, hazard ratio: the risk of death over time in treated versus untreated patients (1 = no difference; below 1 = lower risk with treatment). Bars to scale

1. Giugliani R et al. Am J Med Genet A 2014;164A:1953–64 (10-year data). 2. Quartel A et al. J Inborn Errors Metab Screen 2018;6:1–6, Fig 1, Table 1 (15-year data).

05

Infusion day

What happens at each weekly infusion and how reactions are handled

In this section

Infusion day

Side effects of each ERT

Managing infusion reactions

Infusion day

1 Before

- Ask about fever, infection or breathing problems
- Weigh the patient for the dose
- Record vital signs

2 Premedicate

- Antihistamine, with or without antipyretic, 30–60 min before
- Give only as prescribed

3 Prepare

- Inspect vials for particles or discolouration
- Dilute in 0.9% sodium chloride
- Rotate gently, never shake

4 Infuse

- Start slowly and step up the rate as per prescribing information
- Monitor vital signs
- Keep oxygen or CPAP at hand if used

5 After

- Observe before discharge
- Document tolerance and any reaction
- Confirm the next infusion date

Check first: a patient with fever or an acute respiratory illness may need the infusion postponed. Ask the treating physician before starting.

CPAP, continuous positive airway pressure

Elosulfase alfa Summary of Product Characteristics (EMA); Galsulfase Summary of Product Characteristics (EMA); Akyol MU et al. Orphanet J Rare Dis 2019;14:137.

Elosulfase alfa: safety and tolerability

Patients assessed

176

aged 5–57 years in the randomised, double-blind, placebo-controlled trial (MOR-004)[1]

Most adverse reactions were infusion reactions

Generally mild or moderate[1]

More frequent in the first 12 weeks of treatment, then less frequent over time[1]

Most common infusion-reaction symptoms[1]

- Headache
- Nausea
- Vomiting
- Pyrexia
- Chills
- Abdominal pain

In ≥10% of treated patients and ≥5% more often than with placebo

Serious infusion reactions[1]

- Anaphylaxis
- Hypersensitivity
- Vomiting

IR, infusion reaction; MOR-004, phase 3 trial of elosulfase alfa in MPS IVA

1. Elosulfase alfa Summary of Product Characteristics.

Galsulfase: phase 3 safety data in numbers

56%

had an infusion-associated reaction

33 of 59 patients treated with galsulfase[1]

98%

developed IgG antibodies

53 of 54 patients tested[1]

Week 1–146

when first reactions appeared

a first reaction can occur even after years of treatment[1]

Most common adverse reactions

Pyrexia · rash · pruritus · urticaria · chills or rigors · nausea · headache · abdominal pain · vomiting · dyspnoea[1]

Serious: laryngeal oedema, apnoea, respiratory distress, angioedema, asthma, anaphylactoid reaction[1]

IgG, immunoglobulin G

1. Galsulfase Summary of Product Characteristics.

Managing infusion reactions

Mild

Signs

Flushing, nausea, fever, chills, itching, headache, irritability

Actions

- Stop or slow the infusion
- Check vital signs
- Inform the physician for medication
- Restart slowly once settled

Moderate

Signs

Chest pain, hives, severe headache, vomiting, diarrhoea, abdominal cramps

Actions

- Stop the infusion; check vital signs
- Inform the physician for medication
- Reassess after 30 minutes; if settled, restart at about half the rate
- Premedicate next time as prescribed

Severe (anaphylaxis)

Signs

Breathing difficulty, wheeze, throat or face swelling, low blood pressure, collapse

Actions

- Stop the infusion immediately
- Call the emergency team and follow the anaphylaxis protocol
- Physician decides whether and how to restart ERT

ERT, enzyme replacement therapy

1. Hendriksz CJ et al. Mol Genet Metab 2016;119:131–43. 2. Galsulfase Summary of Product Characteristics. 3. Harmatz P et al. Mol Genet Metab 2008;94:469–75. See also: Miebach E. Int J Clin Pharmacol Ther 2009;47(Suppl 1):S100–6.

06

Ongoing follow-up

Monitoring, the care team, adherence and what to check at every contact

In this section

Follow-up assessments on ERT

The multidisciplinary team

Supporting adherence

Educator checklist

Warning signs that need escalation

Follow-up assessments on ERT

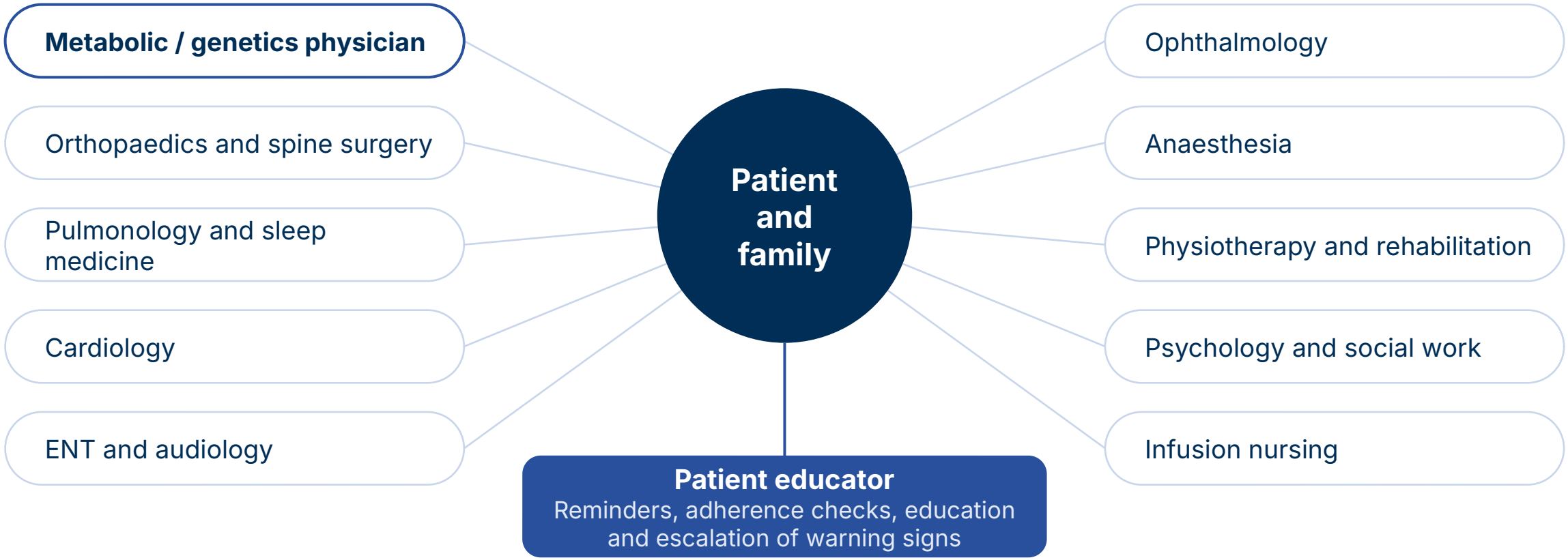
Assessment	What it tracks	MPS IVA (on elosulfase alfa)	MPS VI (on galsulfase)
Cervical spine imaging	Instability and cord compression	X-ray and flexion-extension MRI every 1–3 years	MRI every 2–3 years
Lung function	Breathing capacity (FVC, FEV1, MVV)	Annually until growth stops, then every 2–3 years	Annually
Endurance	Walking capacity	6MWT annually	12MWT every 1–2 years
Heart (ECG, echo)	Valve and heart muscle disease	Every 3 years if normal; annually if involved	Every 1–2 years
Urinary GAGs	Biochemical response to ERT	Urinary KS to track treatment response	Urinary GAGs every 1–2 years
Vision and hearing	Deficits that need support	Regular review	Regular review

Intervals are guide values; the treating physician sets each schedule. Educators remind families of upcoming tests.

6MWT/12MWT, 6-/12-minute walk test; ECG, electrocardiogram; ERT, enzyme replacement therapy; FEV1, forced expiratory volume in 1 s; FVC, forced vital capacity; GAG, glycosaminoglycan; KS, keratan sulfate; MVV, maximum voluntary ventilation

Akyol MU et al. Orphanet J Rare Dis 2019;14:137; Akyol MU et al. Orphanet J Rare Dis 2019;14:118.

The multidisciplinary team



ENT, ear, nose and throat; MPS, mucopolysaccharidosis

Akyol MU et al. Orphanet J Rare Dis 2019;14:137; Akyol MU et al. Orphanet J Rare Dis 2019;14:118.

Supporting adherence: why infusions get missed

44–48%

of MPS VI and IVA patients had ERT interruptions (n=98); drug unavailability was the main cause[1]

63%

of patients on ERT missed at least 1 dose in 6 months, mostly from supply problems or illness[2]

64%

of those in school or work missed a full day every week for infusions[2]

What you can do at every contact

- **Track** infusions given versus scheduled, and ask why any were missed
- **Flag early** supply, transport, cost or paperwork problems to the program team
- **Plan** infusion times around school and work with the family and centre
- **Ill? Call first:** the treating team decides whether to postpone, so don't just skip
- **Reinforce:** every weekly infusion counts; missed doses let GAGs build up again
- **Praise** consistency and never blame

Most missed doses are about access, not motivation. Ask about barriers before you give advice.

ERT, enzyme replacement therapy; GAG, glycosaminoglycan; MPS, mucopolysaccharidosis

1. Solano Villarreal M et al. Mol Genet Metab Rep 2021;28:100769 (Latin America). 2. Yildiz Y, Sivri HS. Turk Arch Pediatr 2021;56:602–9 (Turkey, n=54).

Educator checklist

Adherence

Infusions given vs scheduled; reasons for missed doses such as travel, illness or transport

Tolerance

Any reaction since the last infusion; premedication taken as prescribed

Symptoms

New or worsening problems, especially the five warning signs

Assessments

Upcoming tests and clinic visits; help the family plan and attend

Wellbeing

Pain, sleep, school or work, mood and caregiver strain

Record every contact. Pass any suspected adverse event to pharmacovigilance within the program's reporting timeline, even if it seems minor or unrelated.

Akyol MU et al. Orphanet J Rare Dis 2019;14:137; Akyol MU et al. Orphanet J Rare Dis 2019;14:118; Lampe C et al. Mol Genet Metab Rep 2019;21:100508.

Warning signs that need escalation

Warning sign	What the patient or family may report	Why it matters
Spinal cord compression	Neck pain, new weakness, numbness or tingling, clumsy walking, falls, change in bladder or bowel control	Can cause permanent paralysis; needs urgent assessment
Breathing and sleep	Loud snoring, pauses in breathing, gasping, daytime sleepiness, morning headache	Signals airway obstruction or declining lung function
Heart	Breathlessness on effort, chest pain, palpitations, leg swelling, fainting	May indicate progressing valve disease or heart failure
Planned surgery or sedation	Any operation, dental work or imaging that needs anaesthesia or sedation	High airway and anaesthetic risk; MPS team must plan it
Delayed infusion reaction	Rash, fever, wheeze or swelling in the hours after an infusion	Late-phase reactions need medical review

Escalation rule: report any of these to the treating physician the same day, following the program's escalation procedure.

MPS, mucopolysaccharidosis

Akyol MU et al. Orphanet J Rare Dis 2019;14:137; Akyol MU et al. Orphanet J Rare Dis 2019;14:118; Miebach E. Int J Clin Pharmacol Ther 2009;47(Suppl 1):S100–6.

07

Living with MPS

Safety, daily life and the move to adult care

In this section

Surgery, anaesthesia and emergencies

Living with MPS

Moving to adult care

Key messages for families

Surgery, anaesthesia and emergencies

Situation	What families should do	Why
Planned surgery, dental work or sedated scans	Tell the MPS team early; have it done at a centre experienced in MPS	Narrow, stiff airways make anaesthesia high-risk
Neck and spine, especially in MPS IVA	Keep the neck neutral during procedures, lifting and transfers	An unstable neck can injure the spinal cord
Emergency department visit	Show the MPS alert card and give the treating centre's number	Staff may not know MPS airway and neck risks
Breathing support at home	Bring CPAP or BiPAP to every admission	Keeps the airway open after sedation
After surgery	Report delayed infusions, new weakness or breathing changes	Surgery often interrupts ERT; problems need early review

Alert card: help every family carry one with the diagnosis, ERT, airway and neck risks, and the treating physician's number.

BiPAP, bilevel positive airway pressure; CPAP, continuous positive airway pressure; ERT, enzyme replacement therapy; MPS, mucopolysaccharidosis

Walker R et al. J Inherit Metab Dis 2013;36:211–9; Akyol MU et al. Orphanet J Rare Dis 2019;14:137; Akyol MU et al. Orphanet J Rare Dis 2019;14:118.

Living with MPS: beyond the infusion

Emotional wellbeing

- Diagnosis brings grief, worry and sometimes denial: listen first
- Watch for caregiver exhaustion and low mood
- Offer psychology or social work through the treating team

School and work

- Weekly infusions can cost a day each week: plan schedules early
- Suggest school letters for access, rest and absences
- Encourage independence, hobbies and friendships

Practical support

- Home and school adaptations: ramps, seating, rails
- Mobility, hearing and vision aids as advised
- Link families with MPS patient associations

Cognition is usually normal in MPS IVA and VI: school, work and independent adult life are realistic goals.

MPS, mucopolysaccharidosis

Yildiz Y, Sivri HS. Turk Arch Pediatr 2021;56:602–9; Akyol MU et al. Orphanet J Rare Dis 2019;14:137; Akyol MU et al. Orphanet J Rare Dis 2019;14:118.

Moving from paediatric to adult care

1 ~12 years

- Start talking about adult care
- Name it as a normal step

2 12–14

- Teen learns their diagnosis, ERT and warning signs

3 14–16

- Sees the doctor alone for part of the visit
- Books own appointments

4 16–18

- Adult team named
- Joint clinic if possible
- Medical summary ready

5 18+

- First adult visits attended
- ERT continues with no gaps

Gaps happen around the transfer. In the first adult year, check ERT continuity and visit attendance closely.

ERT, enzyme replacement therapy. Ages are a guide; the treating teams set the timing

Lampe C et al. Transition of patients with mucopolysaccharidosis from paediatric to adult care. Mol Genet Metab Rep 2019;21:100508.

Key messages for families

1. **“Every weekly infusion counts.”**
   The effect of ERT is temporary, so missed doses let GAGs build up again
2. **“ERT slows MPS; it does not undo damage.”**
   Set realistic expectations from the start
3. **“Tell us early.”**
   Report warning signs and reactions without waiting for the next visit
4. **“Mention MPS before any procedure.”**
   Doctors, dentists and anaesthetists must know; carry an alert card
5. **“Keep every check-up.”**
   Regular tests show how the disease and treatment are going
6. **“Plan early for adult care.”**
   Preparation for transition starts around age 12

Stay within your role

- Refer clinical questions to the treating physician
- Never change doses, rates or premedication
- Discuss only approved uses of medicines
- Do not promise specific outcomes
- Use program-approved materials only

ERT, enzyme replacement therapy; GAG, glycosaminoglycan; MPS, mucopolysaccharidosis

Concolino D et al. Ital J Pediatr 2018;44(Suppl 2):120; Akyol MU et al. Orphanet J Rare Dis 2019;14:137; Akyol MU et al. Orphanet J Rare Dis 2019;14:118; Lampe C et al. Mol Genet Metab Rep 2019;21:100508.

08

Pharmacovigilance

Reporting adverse events, special situations and product complaints

In this section

- Why pharmacovigilance matters
- Adverse events and special situations
- Your role and what to collect
- Report or not? Three scenarios
- Product complaints

Why pharmacovigilance matters

Why we collect adverse event and special situation reports

To keep evaluating the product's safety (signal detection): identify, evaluate, communicate and manage new safety information as quickly as possible

1
Collect
Every source counts: patients and caregivers, HCPs, medical information, support programmes, field staff, registries, literature and social media

2
Review
All safety data are used for signal detection, looking for changes in the risk–benefit profile

3
Act
If a new concern emerges: communicate to physicians (Dear Doctor letters) and update the prescribing information (label)

HCP, healthcare professional

Source: BioMarin adverse event identification and reporting training.

Adverse events and special situation events

Adverse event (AE)

Any untoward medical occurrence in a patient given a medicinal product, whether or not it is considered related to the product: an unfavourable sign, symptom or disease linked in time to its use

Special situation event (SSE)

A situation that is not necessarily linked to an adverse event but must still be reported to support continuous monitoring of patient safety

Special situation events to report

Overdose*
Dose above the maximum recommended

Off-label use*
Use for a purpose not in the product information

Misuse or abuse*
Intentional, inappropriate or excessive use with harmful effects

Occupational exposure*
Exposure through one's job

Infection transmission
Infection suspected from a contaminated product

Product complaint*
Alleged problem with quality, label, stability, purity, appearance or effectiveness

Lack of effect*
Product does not produce the expected effect

Medication error*
Error in prescribing, dispensing or giving the product

Interaction
Drug–drug or drug–food interaction changing the effect

Pregnancy or breastfeeding*
Exposure at conception, during pregnancy, delivery or breastfeeding

**Reportable even if not associated with an AE. AE, adverse event; SSE, special situation event*

Source: BioMarin adverse event identification and reporting training.

Your role: report within 1 business day

1 business day

to forward any AE or special situation event to BioMarin Pharmacovigilance once you become aware of it

Record the date you became aware of the event (the initial receipt date)

Four key data points to collect

1 An identifiable reporter
Name and occupation, address, phone, email or other contact details if available

2 An identifiable patient
Age, date of birth, age group, initials and/or gender

3 The suspected product
BioMarin product name, dose, treatment duration, indication and lot number if available

4 At least one adverse event
A brief summary of the event

AE, adverse event

Source: BioMarin adverse event identification and reporting training.

Report or not? Three scenarios

Scenario 1

Pregnancy

A patient has just found out she is pregnant and asks whether it is safe to continue her BioMarin product

REPORT

Pregnancy, with or without an AE, is a special situation event

Scenario 2

Missed infusion

A mother tells you her daughter on a BioMarin product was hospitalised with bronchitis and will miss her weekly infusion

REPORT

The patient had an adverse event while on a BioMarin product

Scenario 3

Infusion reactions

A patient on a BioMarin product has coughing, breathing difficulty, nausea and vomiting at every infusion and is thinking of stopping

REPORT

Infusion reactions are adverse events

AE, adverse event. "BioMarin product" is the product the patient was prescribed. Click to reveal each answer

Source: BioMarin adverse event identification and reporting training.

Product complaints

A product complaint is any information expressing dissatisfaction that alleges a deficiency in a drug product's identity, quality, durability, labelling, purity, stability, appearance, performance, effectiveness, safety or design after release for distribution

Particles in solution

Container or closure

- Bent needle
- Syringe malfunction

Appearance

- Discolouration
- Damaged product or packaging

Labelling

- Missing, unreadable or damaged label

Product

- Extra
- Missing
- Tampered

1 Received

QAPC receives the complaint by fax, phone or email and gives it a unique tracking number

2 Assessed

QAPC assesses it, alerts Pharmacovigilance if an AE may be involved, and acknowledges receipt

3 Investigated

Cross-functional review: QA lot release, QC testing, Pharmacovigilance, clinicians; trends reviewed quarterly

AE, adverse event; QA, quality assurance; QAPC, Quality Assurance Product Complaints; QC, quality control

Source: BioMarin product complaint training.

Reporting a product complaint

5 business days

to notify BioMarin of a product complaint

productcomplaint@bmrn.com

1 Document the details

- Nature of the complaint
- Complainant and contact information
- Product information, including lot number and storage conditions if available

2 Notify BioMarin

- Within 5 business days, by email to productcomplaint@bmrn.com

3 Support the investigation

- Coordinated by BioMarin QAPC
- Document findings, keep records and send copies to BioMarin
- Train employees and agents on complaint reporting

QAPC, Quality Assurance Product Complaints

Source: BioMarin product complaint training.

References

1. Akyol MU et al. Recommendations for the management of MPS IVA. Orphanet J Rare Dis 2019;14:137.
2. Akyol MU et al. Recommendations for the management of MPS VI. Orphanet J Rare Dis 2019;14:118.
3. Burton BK et al. Mol Genet Metab 2025;145:109135.
4. Clarke L et al. J Inborn Errors Metab Screen 2018;6:1–12.
5. Concolino D et al. Ital J Pediatr 2018;44(Suppl 2):120.
6. D'Avanzo F et al. Int J Mol Sci 2021;22:13456.
7. Elosulfase alfa Summary of Product Characteristics.
8. Galsulfase Summary of Product Characteristics.
9. Giugliani R et al. Pediatrics 2007;120:405–18.
10. Gökdoğan C et al. Braz J Otorhinolaryngol 2016;82:281–4.
11. Harmatz P et al. J Pediatr 2006;148:533–9.
12. Harmatz P et al. Mol Genet Metab 2008;94:469–75.
13. Hendriksz CJ et al. Mol Genet Metab 2013;110:54–64.
14. Hendriksz CJ et al. J Inherit Metab Dis 2014;37:979–90.
15. Hendriksz CJ et al. Am J Med Genet A 2015;167A:11–25.
16. Hendriksz CJ et al. Mol Genet Metab 2016;119:131–43.
17. Lampe C et al. Mol Genet Metab Rep 2019;21:100508.
18. McGill JJ et al. Clin Genet 2010;77:492–8.
19. Miebach E. Int J Clin Pharmacol Ther 2009;47(Suppl 1):S100–6.
20. Mitchell J et al. Presenting signs and symptoms of MPS. WORLDSymposium 2018, poster 244.
21. Mitchell JJ et al. Morquio A Registry Study (MARS) 10-year data. WORLDSymposium 2025, poster.
22. Muenzer J. Rheumatology (Oxford) 2011;50(Suppl 5):v4–12.
23. Neufeld EF, Muenzer J. In: The Metabolic and Molecular Bases of Inherited Disease. 8th ed. McGraw-Hill; 2001:3421–52.

References (continued)

24. ABPI. Guidance notes for patient safety and pharmacovigilance in patient support programmes.
25. EMA. Guideline on good pharmacovigilance practices (GVP), Module VI.
26. Lampe C et al. Transition of patients with mucopolysaccharidosis from paediatric to adult care. Mol Genet Metab Rep 2019;21:100508.
27. Solano Villarreal M et al. Mol Genet Metab Rep 2021;28:100769.
28. Walker R et al. Anaesthesia and airway management in mucopolysaccharidosis. J Inherit Metab Dis 2013;36:211–9.
29. Wood T et al. Mol Genet Metab 2012;106:73–82.
30. Yildiz Y, Sivri HS. Difficulties associated with enzyme replacement therapy for mucopolysaccharidoses. Turk Arch Pediatr 2021;56:602–9.

For reference

Appendix

Reference material for questions and further reading

In this section

Glossary of terms

Diagnosis and laboratory tests

Biology of MPS

Skeletal and radiology detail

SmPC dosing and infusion detail

Clinical trial evidence

Glossary

6MWT / 12MWT	Distance walked in 6 or 12 minutes; a measure of endurance
ARSB	Arylsulfatase B, the enzyme missing in MPS VI
CPAP / BiPAP	Machines that keep the airway open with air pressure during sleep
Dysostosis multiplex	Pattern of bone changes seen on X-ray in MPS
ERT	Enzyme replacement therapy
FVC / FEV1	Air breathed out forcefully: total, and in the first second
GAG	Glycosaminoglycan: long sugar chains stored in MPS
GALNS	N-acetylgalactosamine-6-sulfatase, the enzyme missing in MPS IVA
HSCT	Haematopoietic stem cell transplantation
IAR	Infusion-associated reaction
KS / DS	Keratan sulfate / dermatan sulfate, the GAGs stored in MPS IVA / VI
Lysosome	Compartment of the cell where GAGs are normally broken down
MVV	Maximum voluntary ventilation, a test of breathing capacity
Odontoid hypoplasia	Underdeveloped peg of the second neck bone, causing instability
Pharmacovigilance	Collecting and reporting side effects of medicines
uGAG / uKS	Urinary GAG / keratan sulfate levels

Summary: signs of MPS by specialty

Specialty	Signs to look for	Key point
Paediatrics	Subtle, varied early signs; MPS IVA starts at 1–3 years, MPS VI in early childhood	**Diagnosis takes about 4 years on average**
Orthopaedics and rheumatology	Dysostosis multiplex, kyphosis or scoliosis, hip dysplasia, claw hands, short stature	**Hypermobile joints suggest MPS IVA; stiff joints and carpal tunnel suggest MPS VI**
ENT and pulmonology	Recurrent ENT infection, enlarged tonsils, hearing loss, sleep apnoea, macroglossia	**Airway problems are often the first sign and worsen over time**
Cardiology and ophthalmology	Thickened mitral or aortic valves; corneal clouding, optic nerve disease, glaucoma	**Heart involved in 60–100% of patients**
Gastroenterology	Umbilical and inguinal hernias, organomegaly, constipation or diarrhoea	**Hepatosplenomegaly is typical of MPS VI**

ENT, ear, nose and throat; MPS, mucopolysaccharidosis. Macroglossia, enlarged tongue. Sources as cited on the diagnosis appendix slides

Barriers to diagnosing MPS

01

Low disease awareness[2]

Few clinicians have seen MPS

02

Hard to tell MPS types apart[2]

Types overlap in their signs

03

Variable symptoms[3]

Onset and severity differ widely between patients

04

Resembles other disorders[4]

Often mistaken for common rheumatological or skeletal conditions

Knowing how MPS first presents helps clinicians recognise possible cases early, reach a timely diagnosis and manage patients well[5]

MPS, mucopolysaccharidosis

1. Muenzer J. Rheumatology (Oxford) 2011;50(Suppl 5):v4–12. 2. Clarke L et al. 13th International Congress of Inborn Errors of Metabolism, 5–8 September 2017, Rio de Janeiro, Brazil, 649. 3. Giugliani R et al. Pediatrics 2007;120:405–18. 4. Morishita K, Petty RE. Rheumatology (Oxford) 2011;50(Suppl 5):v19–25. 5. Mubarak F et al. 13th International Congress of Inborn Errors of Metabolism, 5–8 September 2017, Rio de Janeiro, Brazil, 662.

Screening and direct tests for MPS

Screening tests

Urinary GAG analysis[1,2]

Accuracy varies by technique; confirm results with a direct test

Enzyme activity analysis (DBS)[1,2]

Easier to collect and transport than whole blood, but can give false positives

Direct tests*

Enzyme activity analysis (leukocytes or fibroblasts)[2]

The gold-standard diagnostic test for MPS

Molecular testing[2]

Can confirm an MPS diagnosis; consult a geneticist

**Direct tests are laboratory tests that can confirm an MPS diagnosis. DBS, dried blood spot; GAG, glycosaminoglycan; MPS, mucopolysaccharidosis*

1. Wood TC et al. J Inherit Metab Dis 2013;36:293–307. 2. Wood T et al. Mol Genet Metab 2012;106:73–82.

Urinary GAG analysis

Sample

Urine[1]

Note: GAG levels are higher in infants and children and fall with age, sometimes into the normal range, which can lead to misreading results[1]

Techniques

- LC-MS/MS[1]
- Quantitative analysis (e.g. DMB)[1]
- Qualitative analysis (e.g. electrophoresis)[1]

Advantages

- MS/MS is increasingly available in laboratories[3]
- Sample collection is non-invasive[1]
- MS/MS is highly accurate and sensitive[2,4]

Limitations

- Not all laboratories offer MS/MS yet[3]
- A positive result must be confirmed by enzyme activity analysis
- High risk of false negatives, especially with quantitative analysis alone

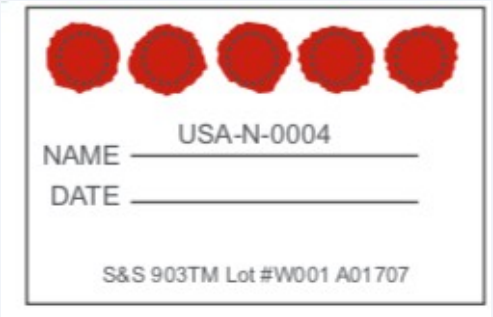
DMB, dimethylmethylene blue; GAG, glycosaminoglycan; LC-MS/MS, liquid chromatography–tandem mass spectrometry

1. Wood T et al. Mol Genet Metab 2012;106:73–82. 2. Kubaski F et al. Mol Genet Metab 2017;120:67–77. 3. BioMarin, data on file. 4. Gelb MH et al. J Inherit Metab Dis 2006;29:397–404.

Dried blood spot (DBS) testing

Sample

Blood collected as dried spots on filter paper



Results

- Complete results in 14–21 days[1]
- Reported by post, phone or fax, or online[1]

Advantages

- Easy to collect, store and transport[1]
- One sample supports several enzyme tests and molecular analysis[2]

Limitations

- Cannot confirm an MPS diagnosis[2]
- Samples are temperature-sensitive; ship without delay[2]
- Repeat the test if results are unclear or samples unusable[1]
- A specialist's clinical evaluation must confirm the diagnosis[1]

DBS, dried blood spot; MPS, mucopolysaccharidosis

1. Lehman TJ et al. Rheumatology (Oxford) 2011;50(Suppl 5):v41–8.

Molecular testing

Sample[1,2]

- Leukocytes from whole blood
- Leukocytes from DBS
- DNA from saliva

Advantages

- Confirms the diagnosis alone when two pathogenic variants are found, one on each allele[3]
- Can confirm the specific MPS type when enzyme results are inconclusive
- Supports carrier and prenatal testing and genetic counselling once the family variant is known[3]

Limitations

- Novel variants or variants of unknown significance need an enzyme test to confirm[2,3]
- Some sequencing methods miss certain variants[2]
- Usually costs more than other tests[3]

DBS, dried blood spot; MPS, mucopolysaccharidosis

1. Lehman TJ et al. Rheumatology (Oxford) 2011;50(Suppl 5):v41–8. 2. Wood TC et al. J Inherit Metab Dis 2013;36:293–307. 3. Wood T et al. Mol Genet Metab 2012;106:73–82. 4. BioMarin, data on file.

Enzyme activity testing in leukocytes or fibroblasts

Sample[1,2]

- Leukocytes isolated from whole blood
- Fibroblasts from skin biopsies

Advantages

- Several enzymes can be tested from one sample[2]
- Allows quicker analysis[1]

Limitations

- Environmental temperatures can compromise leukocyte samples[2]
- Skin punch biopsies are invasive and slow to culture[2]

MPS, mucopolysaccharidosis

1. Lehman TJ et al. Rheumatology (Oxford) 2011;50(Suppl 5):v41–8. 2. Wood TC et al. J Inherit Metab Dis 2013;36:293–307. 3. Wood T et al. Mol Genet Metab 2012;106:73–82.

Prenatal diagnosis

Before pregnancy

Pre-implantation genetic diagnosis

- Genetic profiling of embryos before implantation, sometimes of eggs before fertilisation
- Used like prenatal diagnosis to screen for a specific genetic disease
- **Main advantage:** avoids selective abortion, as the baby is very likely to be free of the disease

Weeks 10–13

Chorionic villus sampling (CVS)

- A sample of tissue is taken from the placenta
- Done earlier than amniocentesis
- Miscarriage risk slightly higher than with amniocentesis

Usually weeks 15–20

Amniocentesis

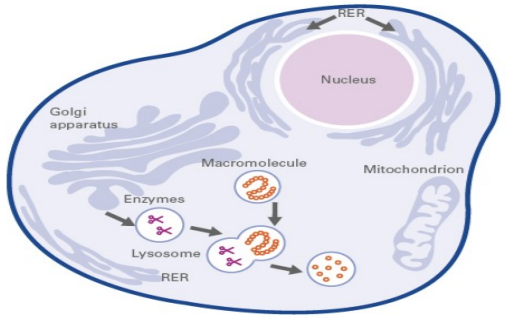
- A thin needle, guided by ultrasound, withdraws a small amount of amniotic fluid
- Can be done up until birth
- Results take 1 day to several weeks; very small risk of pregnancy loss

Prenatal diagnostic tests use fetal or placental cells from CVS or amniocentesis to show whether the fetus has a specific disorder

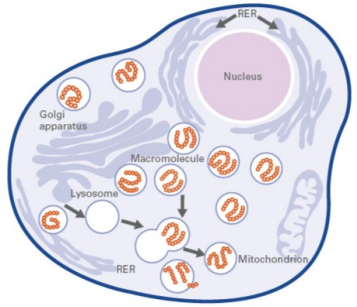
CVS, chorionic villus sampling; PGD, pre-implantation genetic diagnosis

Source: American College of Obstetricians and Gynecologists. Practice Bulletin No. 162: Prenatal diagnostic testing for genetic disorders. Obstet Gynecol 2016;127:e108–22.

Lysosomal storage disorders (LSDs)



Normal cell: macromolecules are degraded by lysosomal enzymes



Lysosomal disorder: accumulation of macromolecules inside the lysosome

What a lysosome does: an organelle whose digestive enzymes break down worn-out cell parts and destroy invading microbes

When lysosomes fail: in most LSDs, undegraded waste cannot be removed from cells[1,2]

~50 LSDs affect the skeleton, brain, skin, heart and nervous system[4]

Batten disease
progressive neurological disorder; a form of NCL[4]

Fabry disease
GLA gene mutations; alpha-galactosidase A is deficient[5]

Gaucher disease (types I–III)
GBA gene mutations; beta-glucocerebrosidase is deficient[6]

Mucopolysaccharidosis (MPS)
one of the enzymes that degrade GAGs is deficient[3]

GAG, glycosaminoglycan; GBA, glucosylceramidase beta; GLA, galactosidase alpha; LSD, lysosomal storage disorder; MPS, mucopolysaccharidosis; NCL, neuronal ceroid lipofuscinosis

1. Staretz-Chacham O et al. Pediatrics 2009;123:1191–207. 2. Heese BA. Semin Pediatr Neurol 2008;15:119–26. 3. Muenzer J. Rheumatology (Oxford) 2011;50(Suppl 5):v4–12. 4. Platt FM et al. Nat Rev Dis Primers 2018;4:27. 5. Germain DP. Orphanet J Rare Dis 2010;5:30. 6. Stirnemann J et al. Int J Mol Sci 2017;18:441.

MPS and glycosaminoglycans (GAGs)

Mucopolysaccharidoses (MPS)

A group of metabolic disorders in which GAGs accumulate because an enzyme that degrades them is deficient

What GAGs are

Long, unbranched polysaccharide chains of repeating disaccharide units, found throughout the body

Components of proteoglycans that regulate many cell processes[1]

Highly polar and water-attracting, so they act as lubricants and shock absorbers

Four GAG chains build up in MPS, each in different tissues[1]

HS	**Heparan sulfate**	Connective tissue; buildup affects the brain
DS	**Dermatan sulfate**	Mucosa and skin
CS	**Chondroitin sulfate**	Skeletal and soft tissues
KS	**Keratan sulfate**	Cornea, cartilage, brain and bone

CS, chondroitin sulfate; DS, dermatan sulfate; GAG, glycosaminoglycan; HS, heparan sulfate; KS, keratan sulfate; MPS, mucopolysaccharidosis

1. Islam T et al. Chemistry, biochemistry, and pharmaceutical potentials of glycosaminoglycans and related saccharides. In: Wong C-H (ed). Carbohydrate-based Drug Discovery. Weinheim: Wiley-VCH; 2003.

MPS types and enzyme deficiencies

Type	Common name	Deficient enzyme	Stored GAG	Hallmark features
I	Hurler, Hurler-Scheie, Scheie	α-L-iduronidase	DS, HS	Skeletal and organ disease; brain in severe form
II	Hunter	Iduronate-2-sulfatase	DS, HS	X-linked, mainly males; similar to MPS I
III	Sanfilippo A–D	Four different enzymes	HS	Mainly brain: behaviour, sleep, regression
IVA	**Morquio A**	**GALNS**	**KS, C6S**	**Severe skeletal, lax joints; normal cognition**
IVB	Morquio B	β-galactosidase	KS	Milder skeletal disease
VI	**Maroteaux-Lamy**	**ARSB**	**DS**	**Skeleton, heart and cornea; normal cognition**
VII	Sly	β-glucuronidase	DS, HS, CS	Very variable; may present before birth
IX	Natowicz	Hyaluronidase	Hyaluronan	Very rare; soft-tissue masses near joints

ARSB, arylsulfatase B; C6S, chondroitin-6-sulfate; CS, chondroitin sulfate; DS, dermatan sulfate; GAG, glycosaminoglycan; GALNS, N-acetylgalactosamine-6-sulfatase; HS, heparan sulfate; KS, keratan sulfate; MPS, mucopolysaccharidosis

Muenzer J. Rheumatology (Oxford) 2011;50(Suppl 5):v4–12; Neufeld EF, Muenzer J. In: The Metabolic and Molecular Bases of Inherited Disease. 8th ed. McGraw-Hill; 2001:3421–52.

Orthopaedics: skeletal involvement in MPS

- Skeletal impairment ranges from **mild platyspondyly** (flattened vertebrae) to **severe deformities** that cause disability and shorten life[1]
- **Short stature** and slowed growth are typical of most MPS types[1]
- **Dysostosis multiplex** (bone abnormalities) is identified on X-ray[2,3]

How dysostosis multiplex develops

- GAGs build up in bone and cartilage cells[4]
- Clusters of GAG-filled cells trigger abnormal apoptosis in the growth plate, disrupting its normal column structure[4]

Apoptosis, programmed cell death, in which the cell is packaged for removal by immune cells, unlike necrosis from injury. GAG, glycosaminoglycan; MPS, mucopolysaccharidosis

1. Neufeld EF, Muenzer J. In: The Metabolic and Molecular Bases of Inherited Disease. 8th ed. McGraw-Hill; 2001:3421–52. 2. Lachman R et al. J Pediatr Rehabil Med 2010;3:109–18. 3. White KW et al. J Pediatr Rehabil Med 2010;3:101–7. 4. Simonaro CM et al. Pediatr Res 2005;57:701–7.

Orthopaedics: general and specialist skeletal features

Pelvis



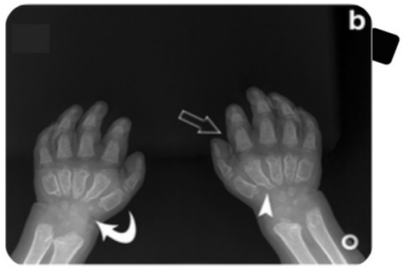
Hip dysplasia[1]
MPS IH (age 8 years)
Rounded iliac wings, inferior tapering of the ilea with a poorly developed acetabulum, underdeveloped medial portion of the proximal femoral epiphysis, increased coxofemoral joint space and coxa valga

Knees



Genu valgum[1]
MPS IH (age 8 years)

Hands



Small, irregular carpal bones, broad and proximally pointed short metacarpals, bullet-shaped phalanges[1]
MPS VI (age 5 years)

Dysostosis multiplex: the set of radiographic changes seen in MPS, here in the pelvis, knees and hands[4]

General features[1–3]

Abnormal gait, bone dysplasia, claw hands, coarse facial features, joint pain, macrocephaly, pectus carinatum, reduced endurance, short stature

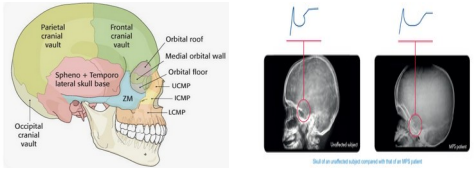
Found on specialist assessment[1–3]

Bone deformities, dysostosis multiplex, genu valgum, joint contractures or laxity without inflammation, spinal subluxation

MPS, mucopolysaccharidosis. Carpus, the wrist bones joining hand to forearm; dysplasia, abnormal growth or development of a tissue or organ; genu valgum, knock-knee

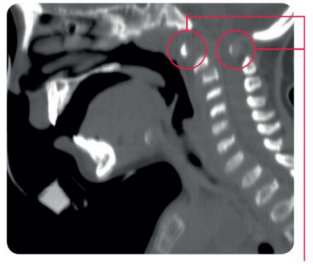
1. Lehman TJ et al. Rheumatology (Oxford) 2011;50(Suppl 5):v41–8. 2. Morishita K, Petty RE. Rheumatology (Oxford) 2011;50(Suppl 5):v19–25. 3. Lachman RS et al. Skeletal Radiol 2014;43:359–69. 4. Palmucci S et al. Insights Imaging 2013;4:443–59.

Radiology: skull, spine and chest



Skull

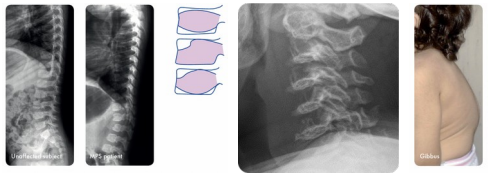
- Macrocephaly, thick skull vault
- Scaphocephaly: long, narrow head
- J-shaped sella turcica, thick diploic space[1]



CAT scan of the cervical spine of an MPS patient depicting the reduction in size of the spinal canal at the craniocervical junction

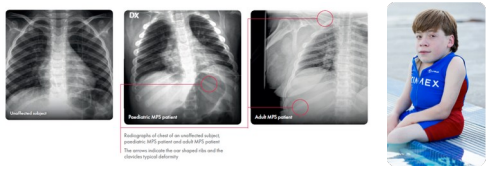
Cervical spine

- Skull-base and atlas–axis deformity[2,3]
- Cervical instability puts the cord at risk[2,3]
- Severe cases need decompression[2]



Vertebral column

- Oval vertebrae, beaked inferiorly[2–4]
- Platyspondyly: flattened vertebrae[4]
- Kyphosis, scoliosis, gibbus; worse in severe MPS[2,3]



Chest

- Oar-shaped ribs are characteristic[2,4]
- Short, thick clavicles[2,4]
- A rigid chest causes breathing problems[2]

MPS, mucopolysaccharidosis. Diploic space, spongy bone between the inner and outer skull tables; gibbus, a sharp spinal hump; macrocephaly, head size above the 98th percentile; platyspondyly, flattened vertebral bodies; sella turcica, the bony saddle at the skull base that holds the pituitary gland

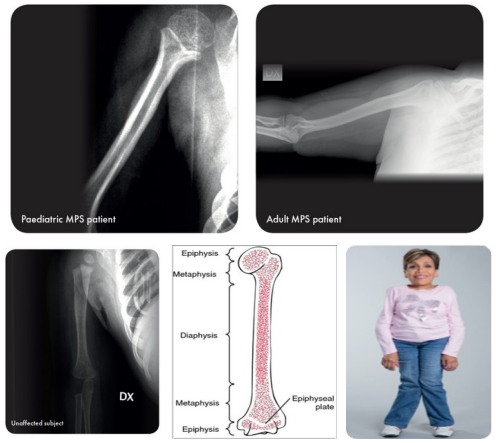
1. Lachman R et al. J Pediatr Rehabil Med 2010;3:109–18. 2. Neufeld EF, Muenzer J. In: The Metabolic and Molecular Bases of Inherited Disease. 8th ed. McGraw-Hill; 2001:3421–52. 3. Beck M et al. J Pediatr Rehabil Med 2010;3:39–46. 4. Lachman RS et al. Skeletal Radiol 2014;43:359–69.

Radiology: hands, long bones and pelvis



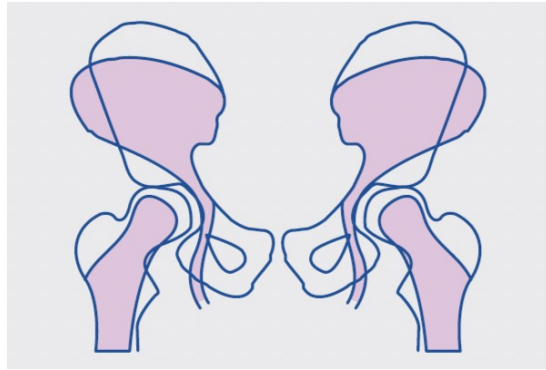
Hands: 'claw hands'

- Short, thick fingers
- Fixed flexion and poor dexterity
- Irregular metaphyseal plates and wide, bullet-shaped phalanges[1]



Long bones and knees

- Mildly hypoplastic epiphyses, displaced growth plates[1]
- Thick, shortened diaphysis[1]
- Bent-knee posture from genu valgum plus spine and pelvis deformity[1,2]



Pelvis

- Hip dysplasia from abnormal bone growth[1]
- Rounded iliac wings[1]
- Orthopaedic surgery is often needed[1]

MPS, mucopolysaccharidosis. Dexterity, skill in using the hands; diaphysis, the shaft of a long bone; epiphysis, the rounded end of a long bone; genu valgum, knock-knee; hypoplastic, underdeveloped; phalanges, the bones of the fingers and toes

1. Lachman R et al. J Pediatr Rehabil Med 2010;3:109–18. 2. Garcia P et al. J Pediatr Rehabil Med 2010;3:63–9.

Orthopaedics: eight signs that point to MPS[1,2]

01
Dysostosis multiplex

02
Kyphosis or scoliosis

03
Vertebral body abnormality
hypoplasia, beaking, platyspondyly, kyphosis/gibbus

04
Bilateral hip dysplasia
or Perthes-like osteonecrosis

05
Claw hands

06
Coarse facial features

07
Joint restriction or laxity
stiffness, or hypermobility

08
Short stature

MPS, mucopolysaccharidosis

1. Lachman RS et al. Skeletal Radiol 2014;43:359–69. 2. Lachman R et al. J Pediatr Rehabil Med 2010;3:109–18.

Rheumatology: joints and short stature in MPS IVA and MPS VI

Joint hypermobility

Specific to MPS IVA[1]

Joint stiffness and contractures

All MPS types except MPS IVA and MPS IX[1]

Carpal tunnel syndrome

May be one of the first signs of MPS VI

Short stature is common in both MPS IVA (Morquio A) and MPS VI (Maroteaux-Lamy)

MPS IVA

Most patients are below the 3rd percentile on the CDC growth chart for children over 2 years (<120 cm)[1]

MPS VI

Growth stalls at about 6–8 years; average adult height is <100 cm in rapidly progressing disease and 140–150 cm in slowly progressing disease[2,3]

CDC, Centers for Disease Control and Prevention; MPS, mucopolysaccharidosis

1. Harmatz P et al. Mol Genet Metab 2013;109:54–61. 2. Giugliani R et al. Genet Mol Biol 2010;33:589–604. 3. Giugliani R et al. Pediatrics 2007;120:405–18. 4. Morishita K, Petty RE. Rheumatology (Oxford) 2011;50(Suppl 5):v19–25.

SmPC: presentation and dosing

	Elosulfase alfa · MPS IVA	Galsulfase · MPS VI
Presentation	1 mg/ml concentrate for solution for infusion; 5 ml vial contains 5 mg elosulfase alfa	1 mg/ml concentrate for solution for infusion; 5 ml vial contains 5 mg galsulfase
Active substance	Recombinant human N-acetylgalactosamine-6-sulfatase (rhGALNS), made in CHO cells	Recombinant human N-acetylgalactosamine-4-sulfatase, made in CHO cells
Appearance	Clear to slightly opalescent, colourless to pale yellow	Clear to slightly opalescent, colourless to pale yellow
Recommended dose	**2 mg/kg** body weight once a week, IV	**1 mg/kg** body weight once a week, IV
Infusion time	3.5 to 4.5 hours, depending on infusion volume	Over 4 hours
Pretreatment	Antihistamines, with or without antipyretics, 30–60 min before infusion	Antihistamines, with or without antipyretics, 30–60 min before infusion

CHO, Chinese hamster ovary; ERT, enzyme replacement therapy; IV, intravenous; SmPC, Summary of Product Characteristics

1. Elosulfase alfa Summary of Product Characteristics. 2. Galsulfase Summary of Product Characteristics.

SmPC: storage, dilution and preparation

	Elosulfase alfa MPS IVA	Galsulfase MPS VI
Storage	2–8 °C; do not freeze or shake; protect from light; no preservatives, so use diluted product within 48 hours	
Supplies	5 ml vials; 0.9% saline bag (100 or 250 ml); low-protein-binding set with 0.2 µm filter	5 ml vials; 0.9% saline bag (100 or 250 ml); syringes for dilution
Bag size	**<25 kg:** 100 ml; **≥25 kg:** 250 ml	**250 ml;** 100 ml if ≤20 kg or at risk of fluid overload

Preparation: the same six steps for both products

01 Inspect: clear to pale yellow; no particles or discolouration

02 Make room: remove saline equal to the concentrate volume

03 Add: slowly add the calculated volume to the bag

04 Mix gently: rotate, do not shake; slight flocculation is acceptable

05 Label: per institution policy; do not mix with other drugs

06 Infuse: through a 0.2 µm in-line filter; discard unused product

1. Elosulfase alfa Summary of Product Characteristics. 2. Galsulfase Summary of Product Characteristics.

SmPC: infusion rates

	Elosulfase alfa	Galsulfase
100 ml bag	**Start:** 3 ml/h **Increase:** to 6 ml/h, then +6 ml/h every 15 min as tolerated **Maximum:** 36 ml/h	**First hour:** 3 ml/h **Then:** up to 39 ml/h for about 3 hours, if well tolerated
250 ml bag	**Start:** 6 ml/h **Increase:** to 12 ml/h, then +12 ml/h every 15 min as tolerated **Maximum:** 72 ml/h	**First hour:** 6 ml/h **Then:** up to 81 ml/h for about 3 hours, if well tolerated
Total duration	At least 3.5 hours (100 ml) or 4.5 hours (250 ml)	About 4 hours

Infusion rates by total infusion volume, per each product's SmPC

1. Elosulfase alfa Summary of Product Characteristics. 2. Galsulfase Summary of Product Characteristics.

SmPC: warnings and special populations

	Elosulfase alfa	Galsulfase
Warnings	Infusion reactions; spinal or cervical cord compression; 8 mg sodium per vial; contains sorbitol (avoid in hereditary fructose intolerance)	Infusion-associated reactions
Pregnancy	No data in pregnant women	Do not use unless clearly necessary
Breast-feeding	Only if benefit outweighs the risk to the infant	Stop breast-feeding during treatment
Fertility	No impairment in nonclinical studies	No impaired fertility or fetal harm in rats and rabbits
Age ≥65 years	Safety and efficacy not established; no alternative dose	Safety and efficacy not established; no alternative dose
Renal or hepatic impairment	–	Not evaluated; no alternative dose
Young children and higher doses	<5 years at 2 mg/kg/week: safety as in older children. Doses up to 4 mg/kg/week: no specific signs	<1 year at 2 mg/kg/week (n=4): safety did not differ meaningfully from 1 mg/kg/week

1. Elosulfase alfa Summary of Product Characteristics. 2. Galsulfase Summary of Product Characteristics.

Dose, dilution and infusion rate by ERT

Elosulfase alfa | MPS IVA

DOSE	2 mg/kg once weekly, IV
VIALS	5 mg in 5 mL; vials = weight × 2 ÷ 5, rounded up
DILUTE	0.9% NaCl: 100 mL if <25 kg; 250 mL if ≥25 kg
RATE <25 KG	3 mL/h for 15 min, then 6 mL/h; increase by 6 mL/h every 15 min to a maximum of 36 mL/h
RATE ≥25 KG	6 mL/h for 15 min, then increase by 12 mL/h every 15 min to a maximum of 72 mL/h
DURATION	About 3.5 to 4.5 hours

Galsulfase | MPS VI

DOSE	1 mg/kg once weekly, IV
VIALS	5 mg in 5 mL; vials = weight ÷ 5, rounded up
DILUTE	0.9% NaCl: 250 mL; 100 mL if <20 kg or at risk of fluid overload
RATE	About 2.5% of the total volume in the first hour, the rest over the next 3 hours
DURATION	At least 4 hours

Both products: remove saline equal to the drug volume, add the drug slowly, rotate gently and never shake; infuse through a 0.2 µm low-protein-binding in-line filter; store vials at 2–8 °C and use the diluted bag promptly.

ERT, enzyme replacement therapy; IV, intravenous; MPS, mucopolysaccharidosis; NaCl, sodium chloride. Always follow the local SmPC and hospital protocol.

1. Elosulfase alfa Summary of Product Characteristics. 2. Galsulfase Summary of Product Characteristics.

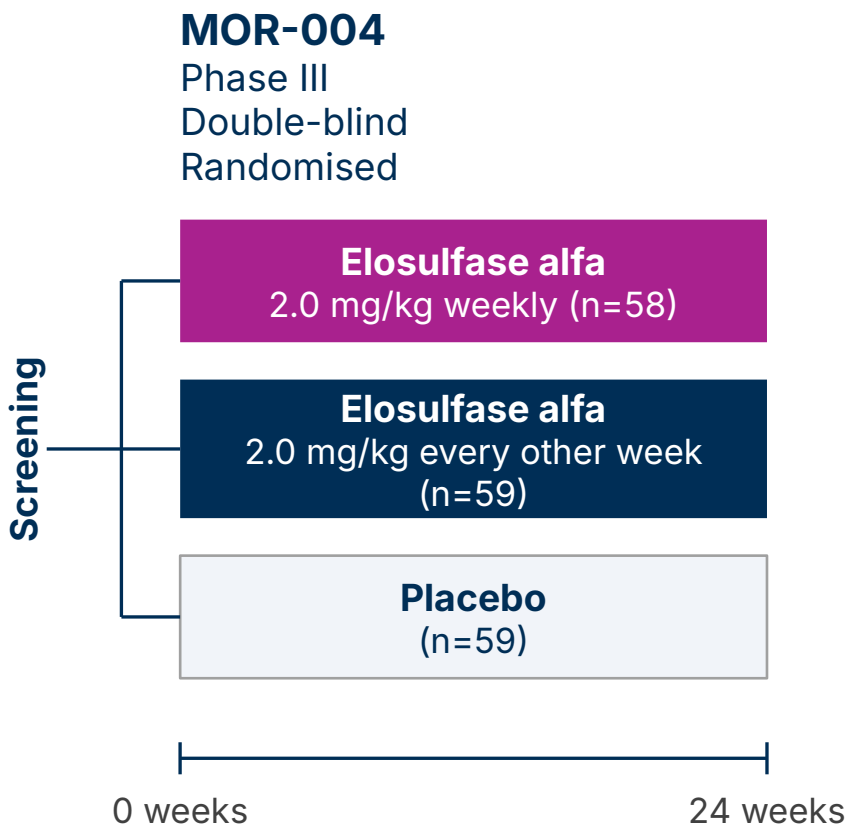
Morquio A clinical programme

Study	Design	n	Start	Duration
MOR-001[1,2]	Observational, longitudinal, untreated natural history (registry): MorCAP	325	2008	Up to 10 years
MOR-002[1,3]	Phase I/II, open-label, dose-escalation: safety and efficacy	20	2010	72 weeks
MOR-100[3]	Extension of MOR-002: long-term safety and efficacy	17	–	192 weeks
MOR-004[1,4]	Phase III, randomised, double-blind, placebo-controlled: efficacy and safety	176	2010	24 weeks
MOR-005[5]	Extension of MOR-004: long-term efficacy and safety	173	–	96 weeks
MOR-006[1]	Phase II, open-label: safety and efficacy in patients with limited ambulation	13	2012	48 weeks
MOR-007[1,6]	Phase II, open-label: safety and efficacy in patients <5 years	15	2011	52 weeks
MOR-008[1,7]	Phase II, randomised, double-blind: safety and physiological effects	25	2012	27 weeks
MARS[1]	Observational, longitudinal (registry)	Recruiting	2014	Up to 10 years

ERT, enzyme replacement therapy (MOR-002 to MOR-008 study elosulfase alfa); MARS, Morquio A Registry Study; MorCAP, Morquio A Clinical Assessment Program

1. ClinicalTrials.gov. 2. Harmatz P et al. Mol Genet Metab 2013;109:54–61. 3. Hendriksz CJ et al. Mol Genet Metab 2012;105:S35. 4. Hendriksz CJ et al. J Inherit Metab Dis 2014;37:979–90. 5. Hendriksz CJ et al. Mol Genet Metab 2016;119:131–43. 6. Jones SA et al. Pediatr Res 2015;78:717–22. 7. Burton BK et al. Am J Med Genet A 2015;167A:2272–81.

MOR-004: study design and results



Study design

- Phase 3, randomised, double-blind, placebo-controlled; 176 patients aged ≥5 years with MPS IVA
- 2 mg/kg weekly, 2 mg/kg every other week, or placebo for 24 weeks
- Primary endpoint: 6MWT; secondary: 3MSCT, urinary KS, lung function

Result

- **Weekly dosing** walked 22.5 m further than placebo at week 24 (P=0.017)
- Every-other-week dosing did not differ from placebo, which is why 2 mg/kg weekly is the approved dose

3MSCT, 3-minute stair climb test; 6MWT, 6-minute walk test; KS, keratan sulfate; MPS, mucopolysaccharidosis

Hendriksz CJ et al. J Inherit Metab Dis 2014;37:979–90.

MOR-004: 6-minute walk test at week 24

Weekly dosing versus placebo

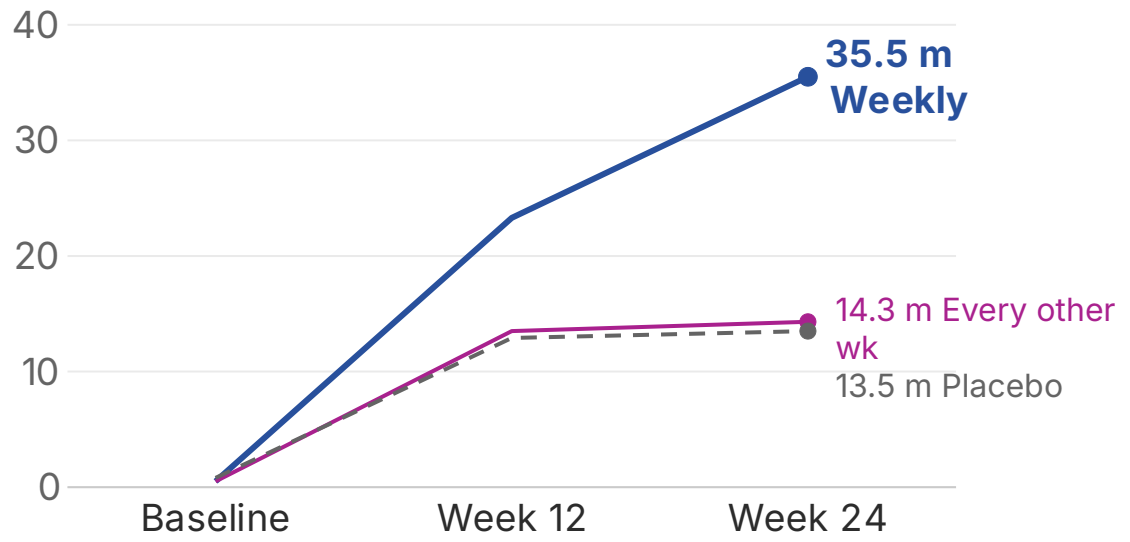
+22.5 m

in the 6MWT at week 24 with elosulfase alfa 2 mg/kg weekly (P=0.0174)[1,2]

Every other week: 2 mg/kg every other week did not differ significantly from placebo[1]

Extension trials: on 2 mg/kg weekly, endurance gains were maintained and urinary KS stayed reduced up to 156 weeks[1]

Change in 6MWT from baseline, metres (mean ± SE)

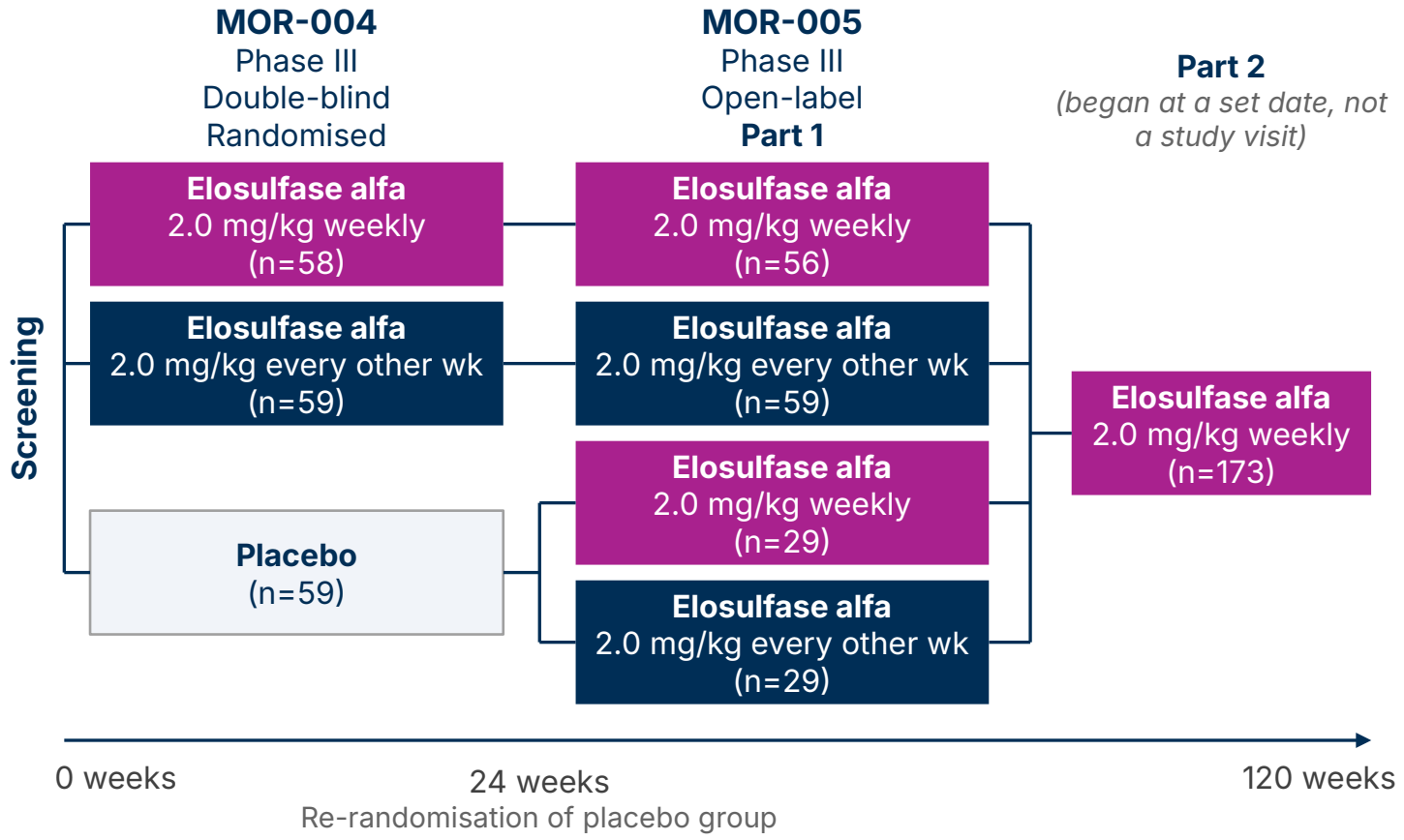


Patients (n)	Baseline	Week 12	Week 24
Weekly	58	58	57
Every other week	59	59	58
Placebo	59	59	59

6MWT, 6-minute walk test: a validated measure of endurance reflecting pulmonary, cardiovascular, circulatory and neuromuscular function. KS, keratan sulfate; SE, standard error

1. Elosulfase alfa Summary of Product Characteristics. 2. Hendriksz CJ et al. J Inherit Metab Dis 2014;37:979–90. 3. ATS Committee on Proficiency Standards. Am J Respir Crit Care Med 2002;166:111–7. 4. McDonald A et al. J Pediatr Rehab Med 2010;3:119–27.

MOR-005: study design



MOR-004, weeks 0–24

176 patients randomised to weekly (n=58), every-other-week (n=59) dosing or placebo (n=59)

MOR-005 part 1

Placebo patients re-randomised to weekly (n=29) or every-other-week (n=29) dosing

MOR-005 part 2

All 173 patients on 2.0 mg/kg weekly; Part 2 began on a set date, not a study visit

Weekly and every-other-week doses are 2.0 mg/kg elosulfase alfa

1. Hendriksz CJ et al. Mol Genet Metab 2016;119:131–43.

MOR-005: populations, efficacy variables and time frame

ITT

All patients from the 24-week phase III study who received at least one dose of elosulfase alfa

MPP

Excluded patients who:

- had orthopaedic surgery during the study (**n=38**)
- missed ≥20% of scheduled infusions (**n=14**)

Efficacy variables

- Safety[1]
- Endurance (6MWT, 3MSCT) and urinary KS[1]
- Respiratory function[2]

Time frame

120 weeks 24 weeks MOR-004 + 96 weeks MOR-005

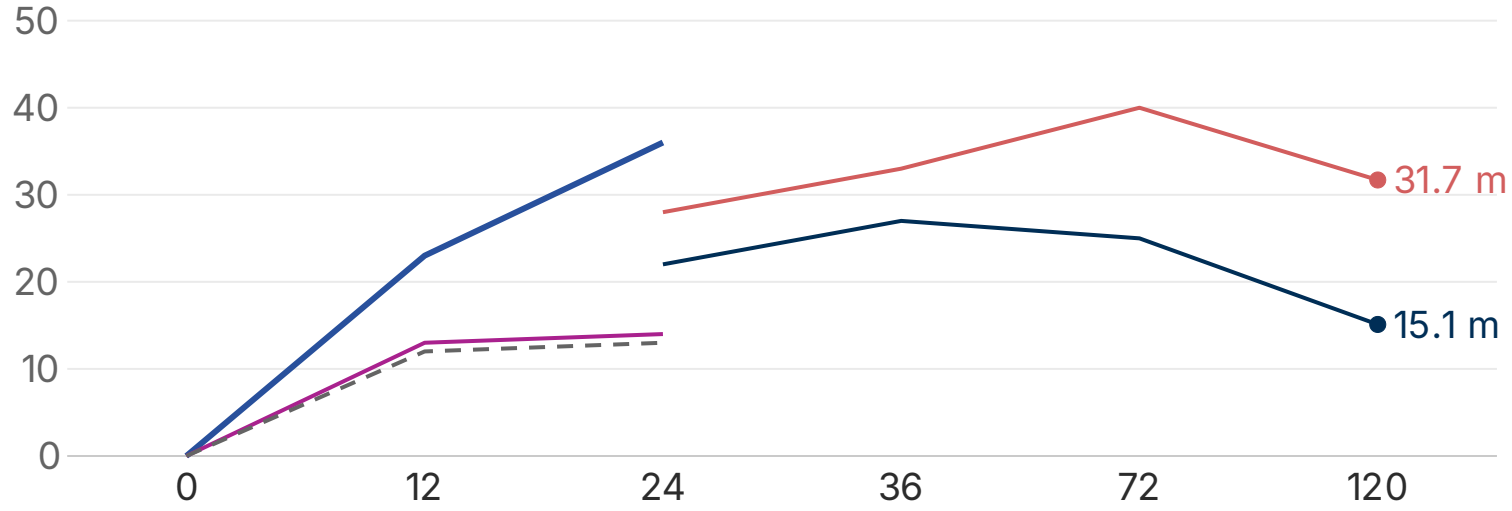
Long-term outcomes were compared with those of untreated patients from MorCAP

3MSCT, 3-minute stair climb test; 6MWT, 6-minute walk test; ITT, intention to treat; KS, keratan sulfate; MorCAP, Morquio A Clinical Assessment Program; MPP, modified per protocol

1. Hendriksz CJ et al. Mol Genet Metab 2016;119:131–43. 2. Hendriksz CJ et al. J Inherit Metab Dis 2016;39:839–47.

MOR-005: 6-minute walk test to week 120

Change in 6MWT from baseline, metres (mean) weeks 0–24 MOR-004, 24–120 MOR-005



Patients (n), week	0	12	24	36	72	120
MOR-004 weekly	58	58	57	–	–	–
MOR-004 every other week	59	59	58	–	–	–
MOR-004 placebo	59	59	59	–	–	–
MOR-005 MPP pooled	–	–	123	122	123	117
MOR-005 ITT pooled	–	–	172	168	166	154

+15.1 m

ITT: mean 6MWT increase from baseline to week 120

+31.7 m

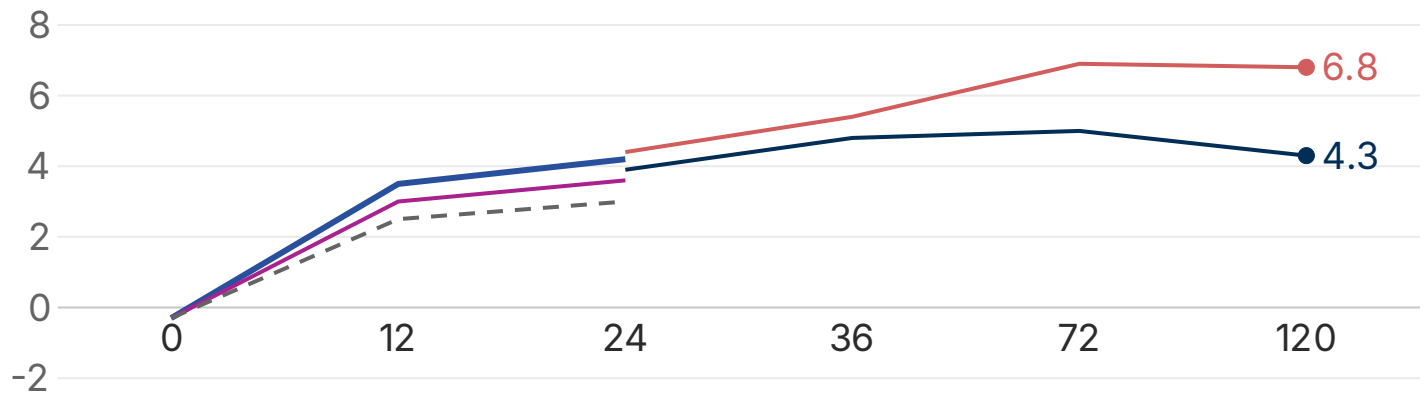
MPP: mean 6MWT increase from baseline to week 120

6MWT, 6-minute walk test; ITT, intention to treat; MPP, modified per protocol; pooled, weekly and every-other-week groups combined. No P values given

1. Hendriksz CJ et al. Mol Genet Metab 2016;119:131–43.

MOR-005: 3-minute stair climb test to week 120

Change in 3MSCT from baseline, stairs/min (mean) weeks 0–24 MOR-004, 24–120 MOR-005



Patients (n), week	0	12	24	36	72	120
MOR-004 weekly	58	58	57	–	–	–
MOR-004 every other week	59	59	58	–	–	–
MOR-004 placebo	59	59	59	–	–	–
MOR-005 MPP pooled	–	–	123	122	123	117
MOR-005 ITT pooled	–	–	172	168	166	154

+4.3

stairs/min, ITT: mean 3MSCT increase from baseline to week 120

+6.8

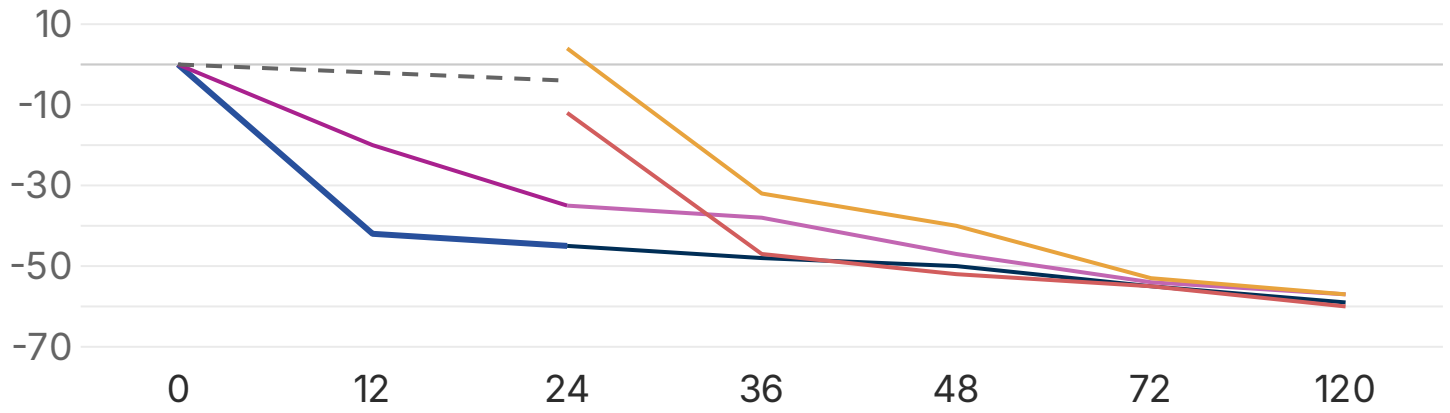
stairs/min, MPP: mean 3MSCT increase from baseline to week 120

3MSCT, 3-minute stair climb test; ITT, intention to treat; MPP, modified per protocol; pooled, weekly and every-other-week groups combined. No P values given

1. Hendriksz CJ et al. Mol Genet Metab 2016;119:131–43.

MOR-005: urinary KS to week 120

Change in urinary KS from baseline, % (mean) weeks 0–24 MOR-004, 24–120 MOR-005



Patients (n), week	0	12	24	36	48	72	120
MOR-004 weekly	58	55	54	–	–	–	–
MOR-004 every other week	59	58	57	–	–	–	–
MOR-004 placebo	58	56	55	–	–	–	–
MOR-005 weekly → weekly	–	–	53	52	52	51	47
MOR-005 every other wk → every other wk	–	–	57	56	56	56	49
MOR-005 placebo → weekly	–	–	28	28	27	28	24
MOR-005 placebo → every other wk	–	–	26	27	28	25	22

–59.4%

ITT: mean reduction in urinary KS from baseline to week 120

–62.3%

MPP: mean reduction in urinary KS from baseline to week 120

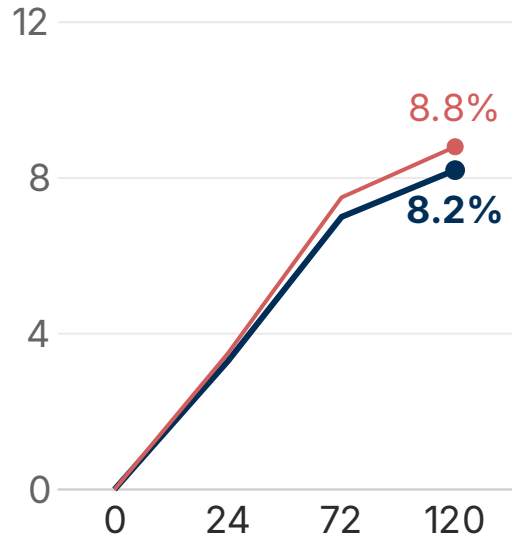
ITT, intention to treat; KS, keratan sulfate; MPP, modified per protocol. MOR-005 groups named by MOR-004 dose → MOR-005 dose. No P values given

1. Hendriksz CJ et al. Mol Genet Metab 2016;119:131–43.

MOR-005: lung function to week 120

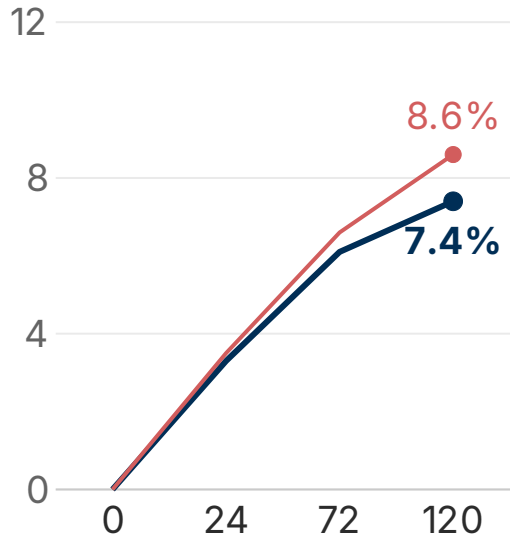
Mean % change from baseline, MOR-004/005 combined study week

FVC



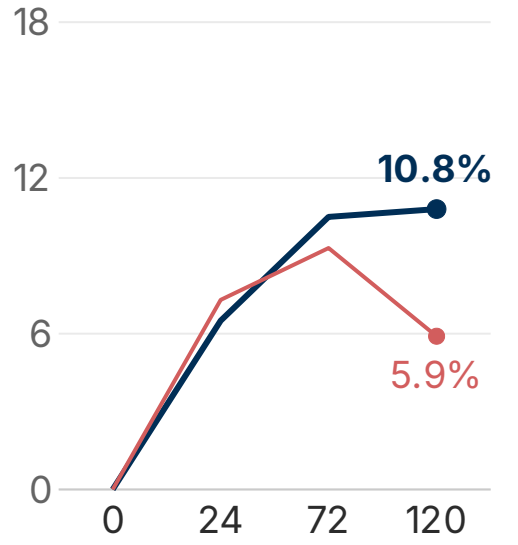
n	0	24	72	120
ITT	162	160	157	142
MPP	118	116	118	112

FEV1



n	0	24	72	120
ITT	161	158	156	142
MPP	118	116	118	113

MVV



n	0	24	72	120
ITT	153	148	144	130
MPP	112	108	111	105

FEV1, forced expiratory volume in 1 second; FVC, forced vital capacity; ITT, intention to treat; MPP, modified per protocol; MVV, maximum voluntary ventilation

1. Hendriksz CJ et al. Mol Genet Metab 2016;119:131–43.

MorCAP comparator population

Untreated patients from the MorCAP study provided baseline data for untreated disease[1]

MorCAP 1

Equivalent to the ITT population

- Morquio A patients aged ≥5 years
- Baseline 6MWT distance 30–325 m
- Longitudinal data available at Year 1 and/or Year 2

MorCAP 2

Equivalent to the MPP population

- As MorCAP 1, but excluding patients who had orthopaedic surgery within 3 months of recruitment or in the following 2 years

Both subpopulations had baseline values comparable with MOR-005[1]

6MWT, 6-minute walk test; ITT, intention to treat; MorCAP, Morquio A Clinical Assessment Program; MPP, modified per protocol

1. Hendriksz CJ et al. J Inherit Metab Dis 2016;39:839–47.

MorCAP (MOR-001): 6-minute walk test in untreated patients

212 m

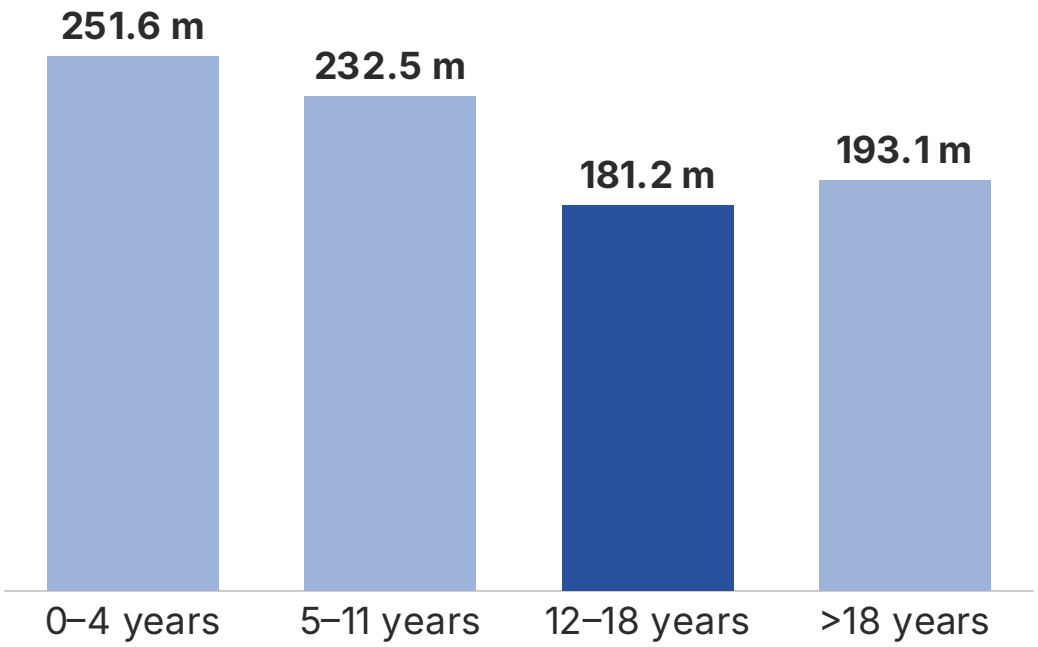
overall mean 6MWT distance (±152), against a lower limit of normal of 470–664 m for ages 4–16 and 500–580 m for adults[1]

–51.3 m

22% mean difference between patients aged 5–11 (n=127) and 12–18 (n=84)[1]

Why 6MWT matters: change in 6MWT reflects improvement or decline in cardiac, respiratory and musculoskeletal function, and in disease progression[1,3]

Baseline 6MWT by age group, metres (mean)



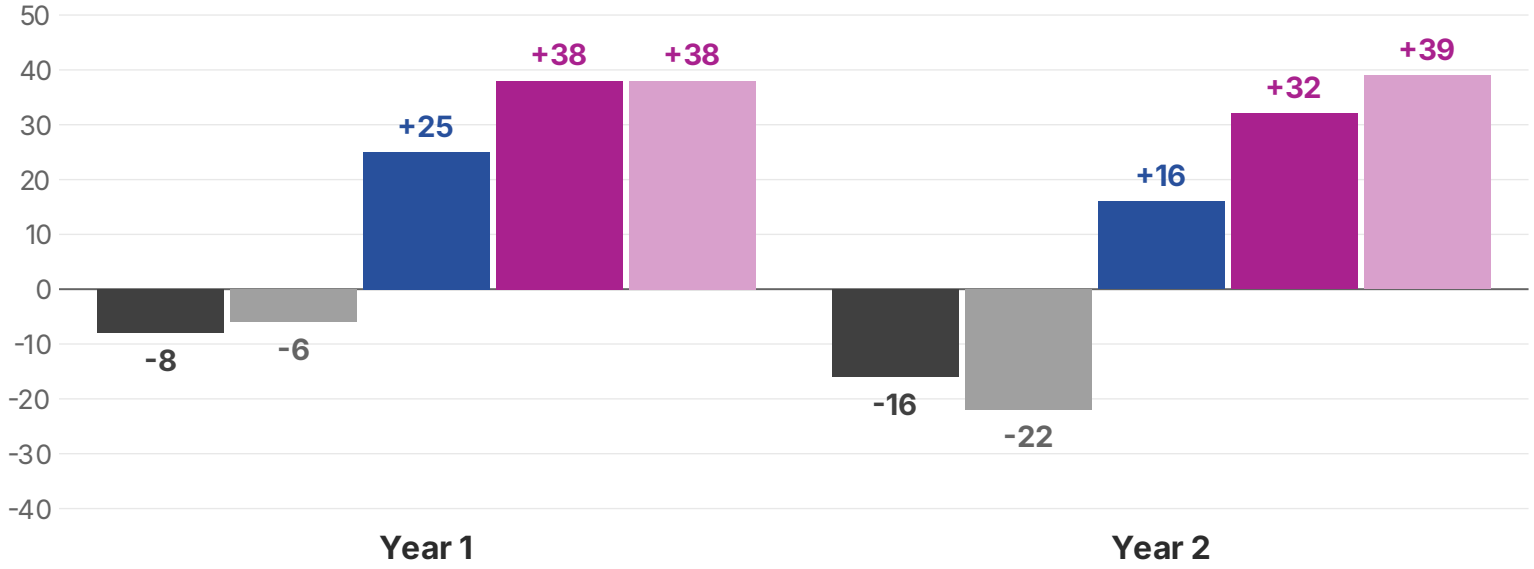
Baseline data from MorCAP: multicentre, multinational, cross-sectional and longitudinal study of 325 patients[1]

6MWT, 6-minute walk test; MorCAP, Morquio A Clinical Assessment Program. Progressive multi-organ involvement sharply reduces endurance before age 18

1. Harmatz P et al. Mol Genet Metab 2013;109:54–61. 2. Tomatsu S et al. Curr Pharm Biotechnol 2011;12:931–45. 3. McDonald A et al. J Pediatr Rehab Med 2010;3:119–27.

6MWT: treated patients versus MorCAP

6MWT LS mean change from baseline, metres



Patients (n)	Year 1	Year 2
MorCAP 1	80	40
MorCAP 2	67	27
ITT pooled	166	154
MPP pooled	123	117
MPP weekly	43	41

Sustained effect

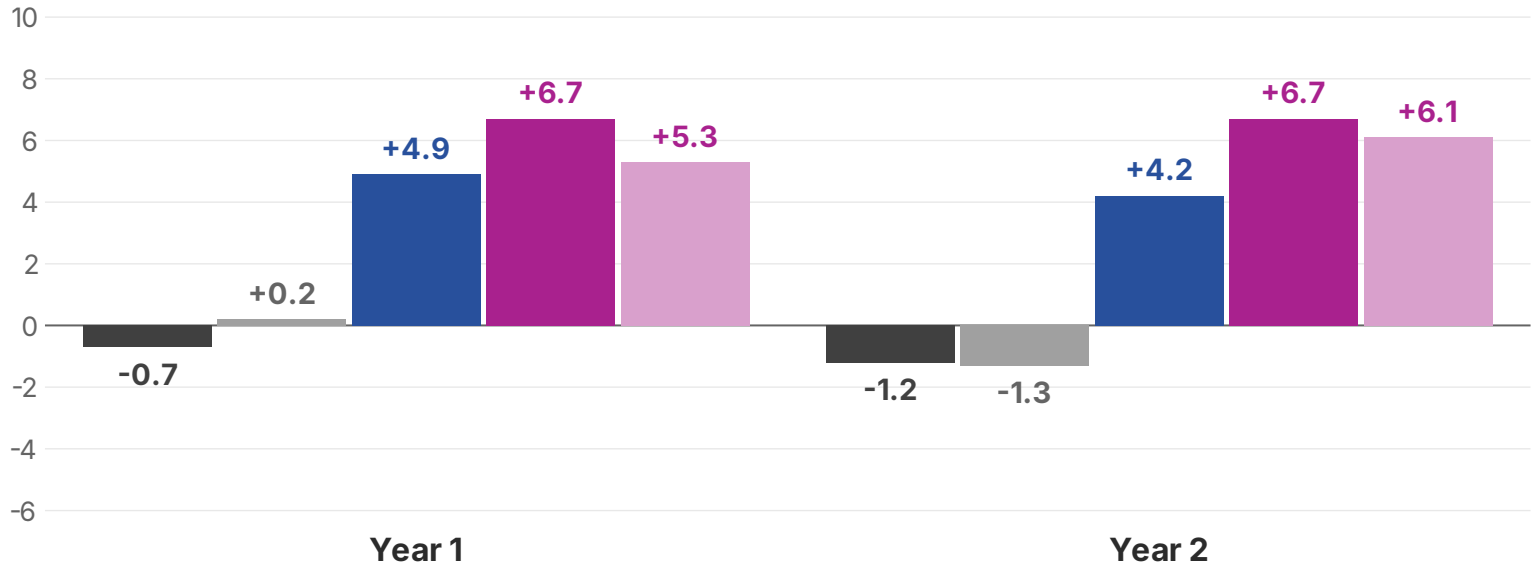
MOR-005 ITT and MPP groups improved 6MWT at both 1 and 2 years (P<0.05)

6MWT, 6-minute walk test; ITT, intention to treat; LS, least squares; MorCAP, Morquio A Clinical Assessment Program; MPP, modified per protocol; QW, weekly

1. Hendriksz CJ et al. Mol Genet Metab 2016;119:131–43.

3MSCT: treated patients versus MorCAP

3MSCT LS mean change from baseline, stairs/min



Patients (n)	Year 1	Year 2
MorCAP 1	80	40
MorCAP 2	67	27
ITT pooled	164	154
MPP pooled	123	117
MPP weekly	43	41

Sustained effect

MOR-005 ITT and MPP groups improved 3MSCT at both 1 and 2 years (P<0.05)

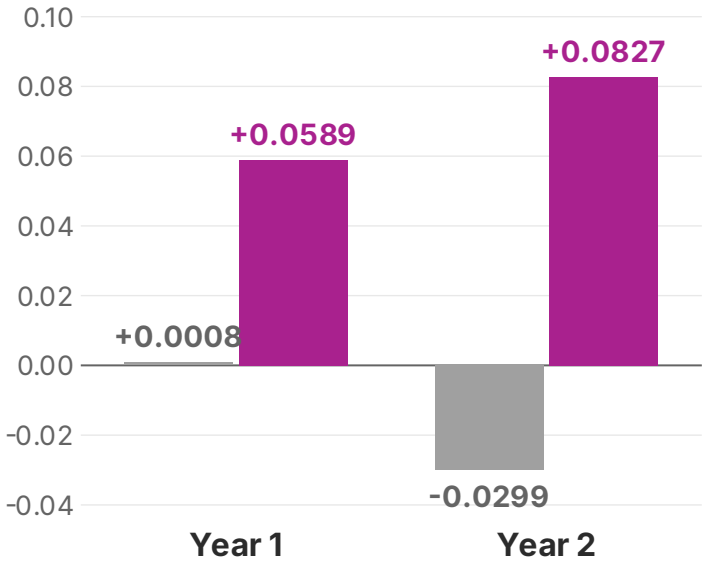
3MSCT, 3-minute stair climb test; ITT, intention to treat; LS, least squares; MorCAP, Morquio A Clinical Assessment Program; MPP, modified per protocol; QW, weekly

1. Hendriksz CJ et al. Mol Genet Metab 2016;119:131–43.

Lung function: treated patients versus MorCAP

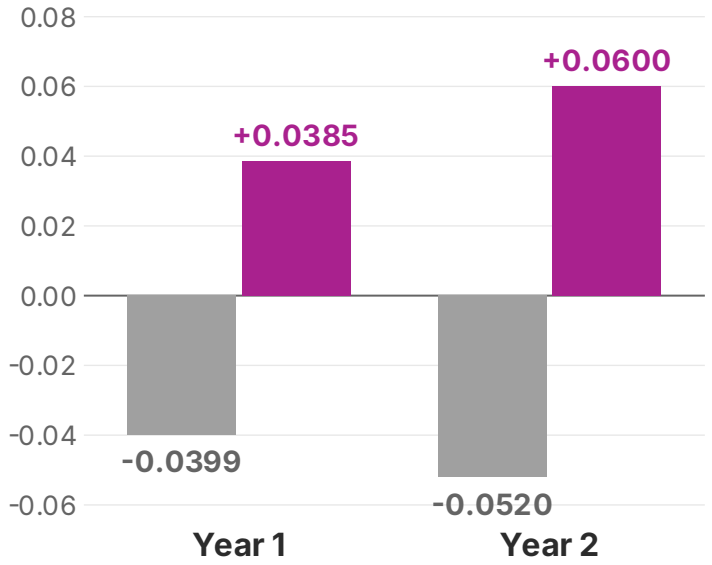
LS mean change from baseline

FVC, L Year 1: P=0.0279 · Year 2: P=0.0429



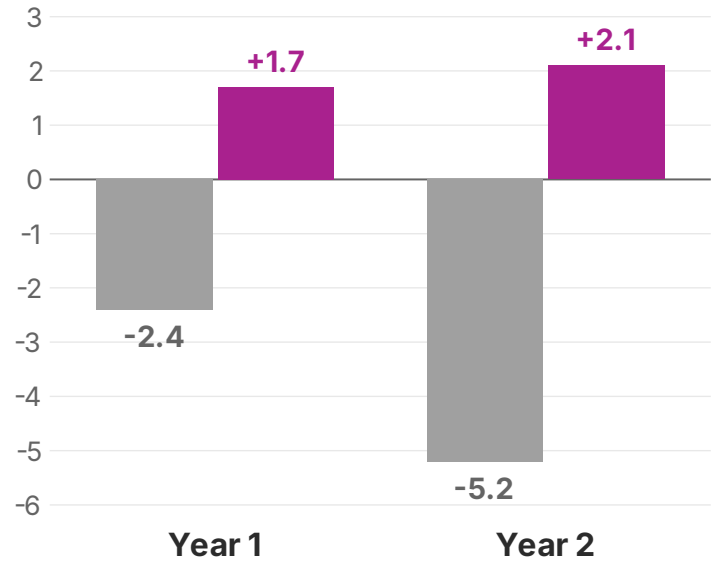
n Untreated 63 / 25 Treated 117 / 111
(Year 1 / Year 2)

FEV1, L Year 1: P=0.0079 · Year 2: P=0.0339



n Untreated 63 / 24 Treated 117 / 112
(Year 1 / Year 2)

MVV, L/min Year 1: P=0.0221 · Year 2: P=0.0127



n Untreated 60 / 22 Treated 110 / 104
(Year 1 / Year 2)

Untreated (MorCAP) **Elosulfase alfa (MOR-004/005), MPP population**

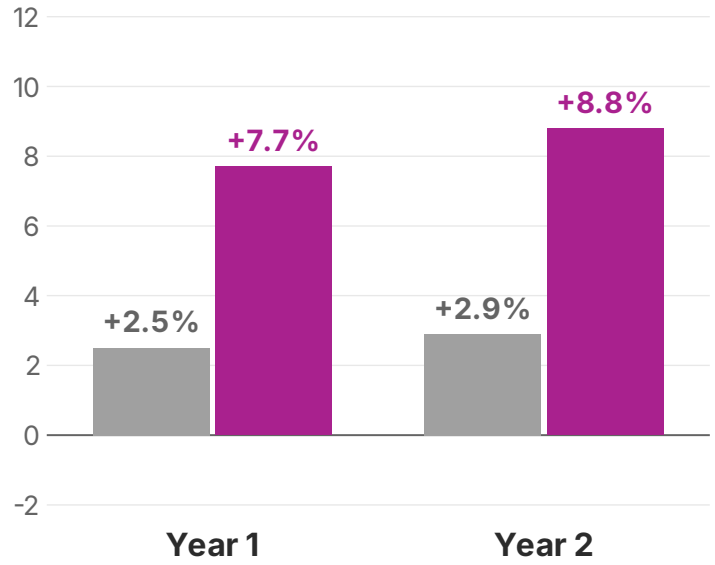
FEV1, forced expiratory volume in 1 second; FVC, forced vital capacity; LS, least squares; MorCAP, Morquio A Clinical Assessment Program; MPP, modified per protocol; MVV, maximum voluntary ventilation

1. Hendriksz CJ et al. J Inherit Metab Dis 2016;39:839–47.

Lung function: percentage change versus MorCAP

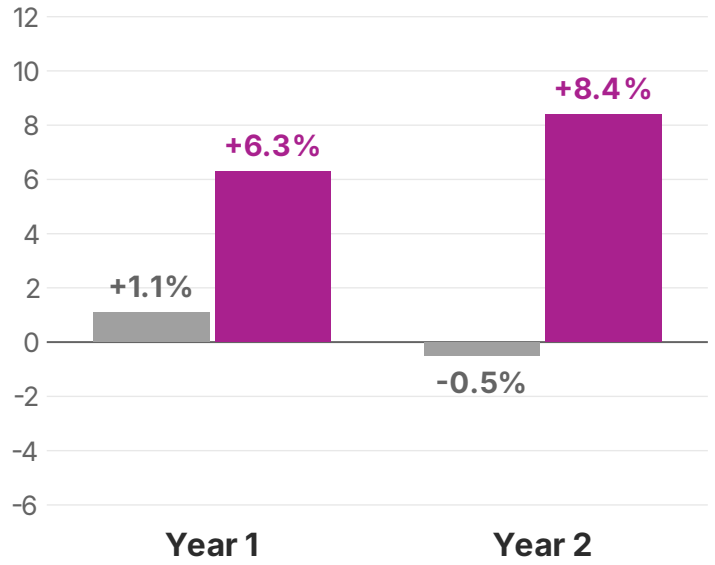
LS mean % change from baseline

FVC Year 1: P=0.0215 · Year 2: NS



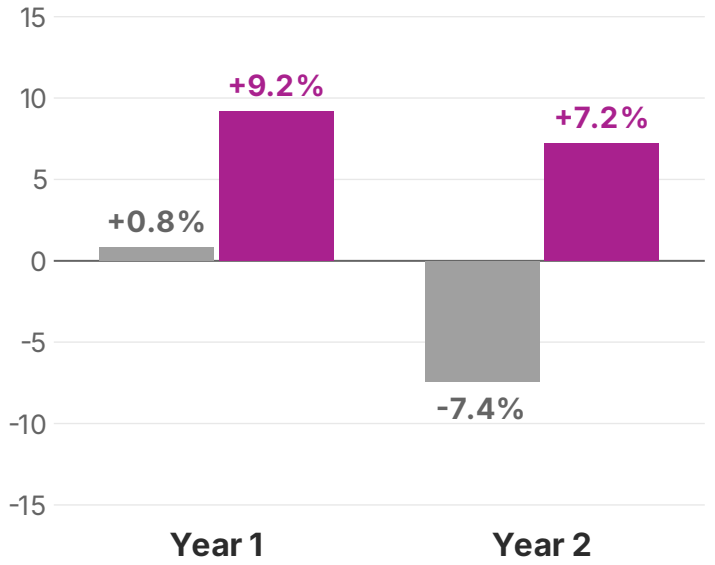
n Untreated 63 / 25 Treated 117 / 111
(Year 1 / Year 2)

FEV1 Year 1: P=0.0443 · Year 2: NS



n Untreated 63 / 24 Treated 117 / 112
(Year 1 / Year 2)

MVV Year 1: NS · Year 2: NS



n Untreated 60 / 22 Treated 110 / 104
(Year 1 / Year 2)

Untreated (MorCAP) **Elosulfase alfa (MOR-004/005), MPP population**

FEV1, forced expiratory volume in 1 second; FVC, forced vital capacity; LS, least squares; MorCAP, Morquio A Clinical Assessment Program; MPP, modified per protocol; MVV, maximum voluntary ventilation; NS, not significant

1. Hendriksz CJ et al. J Inherit Metab Dis 2016;39:839–47.

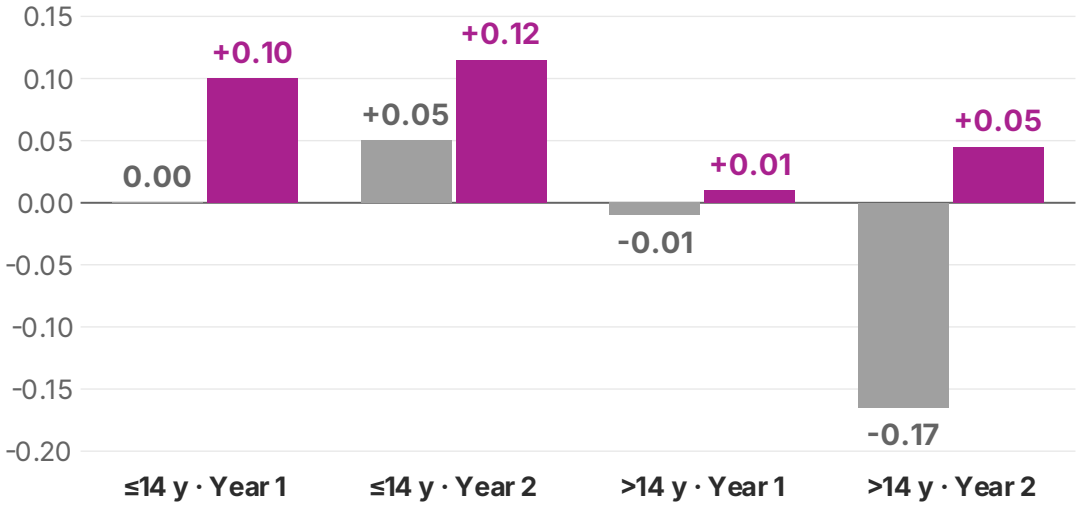
Lung function by age: treated patients versus MorCAP

Aged 14 or under
Both cohorts improved; treated patients improved more

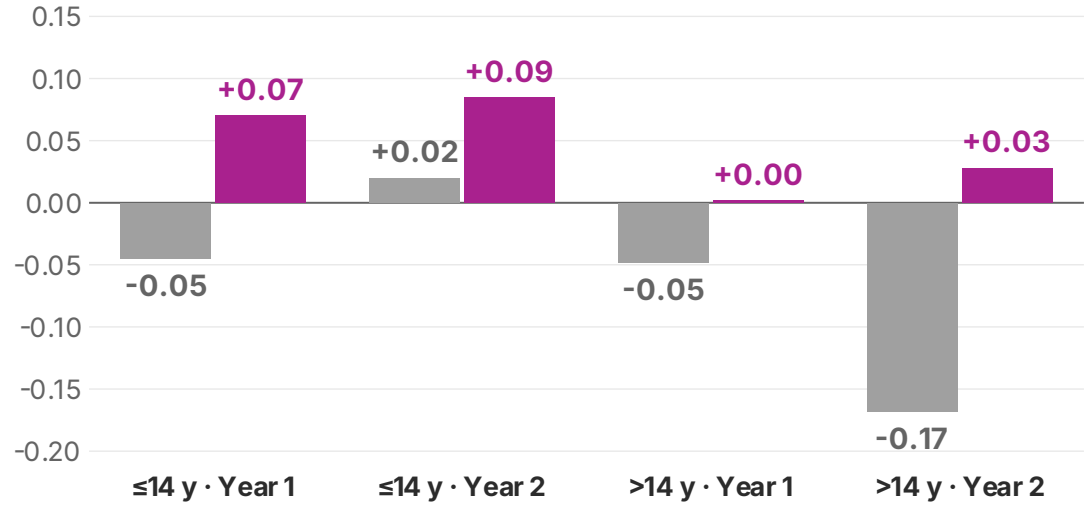
Over 14
Treated patients improved; untreated patients deteriorated

Role of growth
FVC tracked height in younger patients; other factors likely drove gains in older ones

FVC, L ≤14 y: Year 1: P=0.01 · Year 2: NS · >14 y: Year 1: NS · Year 2: P=0.03



FEV1, L ≤14 y: Year 1: P=0.007 · Year 2: NS · >14 y: Year 1: NS · Year 2: P=0.01



Untreated (MorCAP) Elosulfase alfa (MOR-004/005) LS mean change from baseline

FEV1, forced expiratory volume in 1 second; FVC, forced vital capacity; LS, least squares; MorCAP, Morquio A Clinical Assessment Program; NS, not significant

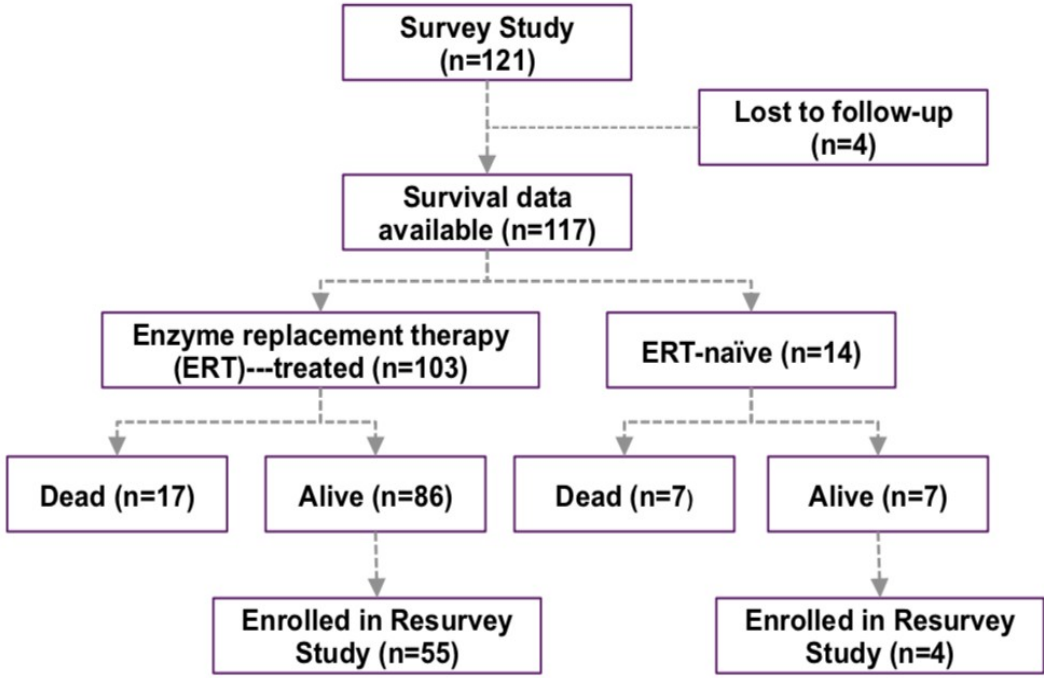
1. Hendriksz CJ et al. J Inherit Metab Dis 2016;39:839–47.

Galsulfase clinical studies

Study	Type	Year	Patients enrolled (completed)	Duration	Refs
Phase 1/2[a]	Randomised, double-blind, dose-comparison / open-label extension	2000–2005	7 (5)	240 weeks	1,2
Phase 2[a]	Open-label	2002–2006	10 (10)	144 weeks	2,3
Phase 3[a]	Randomised, double-blind, placebo-controlled / open-label extension	2003–2006	39 (38)	96 weeks	2,4
Phase 4	Open-label, dose-comparison (age 3.3–12.7 months)	2012	4 (4)	1–3 years	5
CSP	Observational	2005–ongoing	223 (107)[b]	Ongoing	6
Survey Study	Cross-sectional	2002	123 (121)	Baseline	7
Resurvey Study	10-year follow-up of Survey Study	2012	59 (efficacy); 117 (survival)	10-year follow-up	8
Follow-up study	15-year follow-up of Survey Study	2018	121 (111)	15-year follow-up	9

a. Phase 1/2, 2 and 3 trials had open-label extension phases up to 240 weeks. b. Of 223 patients enrolled in CSP, 107 had high- or low-baseline uGAG levels and received treatment for at least 6 months. CSP, Clinical Surveillance Programme; uGAG, urinary glycosaminoglycans

1. Harmatz P et al. J Pediatr 2004;144:574–80. 2. Harmatz P et al. Mol Genet Metab 2008;94:469–75. 3. Harmatz P et al. Pediatrics 2005;115:e681–9. 4. Harmatz P et al. J Pediatr 2006;148:533–9. 5. Harmatz PR et al. J Inherit Metab Dis 2014;37:277–87. 6. Harmatz P et al. J Inherit Metab Dis 2019;1–8. 7. Swiedler SJ et al. Am J Med Genet A 2005;134A:144–50. 8. Giugliani R et al. Am J Med Genet A 2014;164A:1953–64. 9. Quartel A et al. J Inborn Errors Metab Screen 2018;6:1–6.



	ERT group		ERT-naïve group	
Age group	**Baseline**	**Follow-up**	**Baseline**	**Follow-up**
4–12 years	33	0	0	0
13–18 years	10	22	0	0
19–24 years	8	14	1	0
≥25 years	4	19	3	4

Number of patients by age group at baseline and at follow-up

1. Giugliani R et al. Am J Med Genet A 2014;164A:1953–64.

MPS VI Survey Study: urinary GAG levels and disease severity

URINARY GAG LEVEL, µg/mg CREATININE

<100	100–200	>200
low	intermediate	high

Low levels

>20 years

Low uGAG levels were seen in patients over 20 years of age[1]

Higher level, more severe disease

High levels were associated with[1]

- **Age-adjusted short stature**
- **Low body weight**
- **Compromised lung function**
- **Reduced joint range of motion**

6MWT Walk distance was impaired **across all age groups** and **across all uGAG levels**[1]

6MWT, 6-minute walk test; GAG, glycosaminoglycan; MPS, mucopolysaccharidosis; uGAG, urinary GAG

1. Swiedler SJ et al. Am J Med Genet A 2005;134A:144–50.

Survey Study: 15-year follow-up

16.5%

mortality with galsulfase

Patients who received galsulfase at any time, 10-year Resurvey[1]

50%

mortality without treatment

Treatment-naïve patients in the same 10-year Resurvey[1]

12.5%

5-year mortality on galsulfase

Commercial use data, 2005–2016[1]

2001–2002

Survey Study: 121 patients with MPS VI

10 years later

Resurvey: same patients, 10 years on

2005–2016

Commercial-use mortality data

Galsulfase ERT can increase life expectancy in MPS VI over at least 15 years; treated patients also showed physical improvements[1]

ERT, enzyme replacement therapy; MPS, mucopolysaccharidosis

1. Quartel A et al. J Inborn Errors Metab Screen 2018;6:1–6.

Resurvey Study: 10-year follow-up

117
treated and untreated patients followed up[1]

6.8 years
average time on galsulfase ERT[1]

98%
compliance in the clinical studies[2]

95%
compliance in the Resurvey Study[1]

About the study

- 10-year follow-up of medical history, clinical assessment and survival in patients from the original Survey Study[1]
- Results were consistent with clinical trials of long-term galsulfase use in MPS VI[1]

Key message

Start galsulfase as soon as MPS VI is confirmed[1]

Long-term Resurvey data and the phase 3 trial results both reaffirm early treatment and good compliance

ERT, enzyme replacement therapy; MPS, mucopolysaccharidosis

1. Giugliani R et al. Am J Med Genet A 2014;164A:1953–64. 2. Harmatz P et al. Mol Genet Metab 2008;94:469–75.

Resurvey Study: results after about 10 years of galsulfase

Urinary GAGs

−87.9%

reduction in uGAG from a mean 321.3 µg/mg creatinine at baseline

- All treated patients (n=55) reached uGAG <100 µg/mg creatinine[1]
- No P value available

Endurance (6MWT, n=46)

+65.7 m

a mean 21% increase over baseline (P<0.0001)

- All age subgroups improved[1]
- uGAG <100: **+19%** (P=0.0033); 100–200: **+11%**; ≥200: reduced distance[1]

Lung function

+29% FVC

and +18% FEV1 over baseline (P<0.0001; P=0.001)

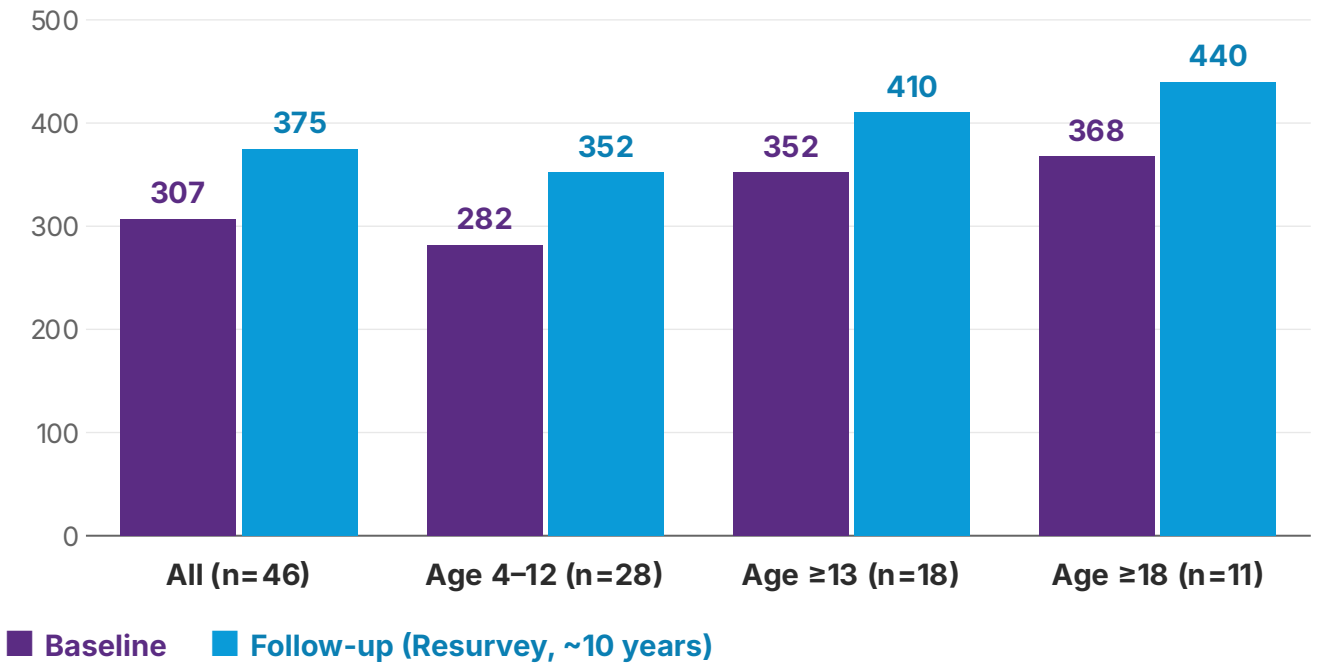
- <13 years: **FVC +68%** (P=0.0002), **FEV1 +55%** (P=0.0005)[1]
- ≥13 years: **FVC +13%** (P=0.011); FEV1 stable[1]

6MWT, 6-minute walk test; FEV1, forced expiratory volume in 1 second; FVC, forced vital capacity; uGAG, urinary glycosaminoglycan. No data from ERT-naïve patients for comparison

1. Giugliani R et al. Am J Med Genet A 2014;164A:1953–64.

Galsulfase: endurance after 10 years

6MWT distance walked by age at baseline, metres (mean)



- Patients who completed the 6MWT (n=46) walked **65.7 m further** than at baseline, a mean increase of 21% (P<0.0001)[1,a–c]
- All age subgroups improved in mean distance walked (P values not available)[1,b]

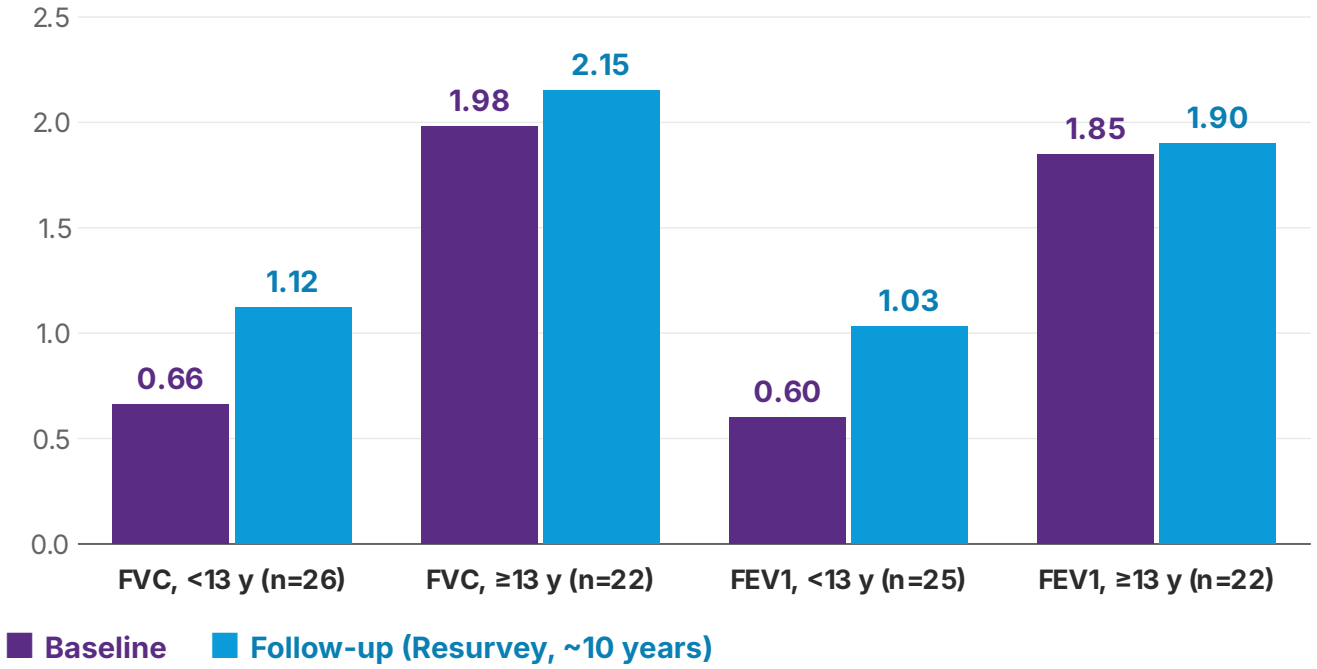
Why endurance matters: it reflects disease change, cardiopulmonary health, exercise capacity, mobility and extent of morbidity[1,2]

a. Patients who completed the 6MWT had been on galsulfase for a mean 6.9 years (±2.2). b. Eight patients could not take part because they were not independently mobile. c. Consistent with the phase 3 trial, whose values came from the 6-minute point of the 12MWT. Chart adapted from Giugliani et al. 2014. 6MWT, 6-minute walk test; 12MWT, 12-minute walk test

1. Giugliani R et al. Am J Med Genet A 2014;164A:1953–64. 2. Morales-Blanhir JE et al. J Bras Pneumol 2011;37:110–7.

Galsulfase: lung function after 10 years

FVC and FEV1 by age at baseline, litres (mean)



Overall

FVC rose 29% (P<0.0001) and **FEV1 18%** (P=0.001) over baseline[1,a]

By age at baseline

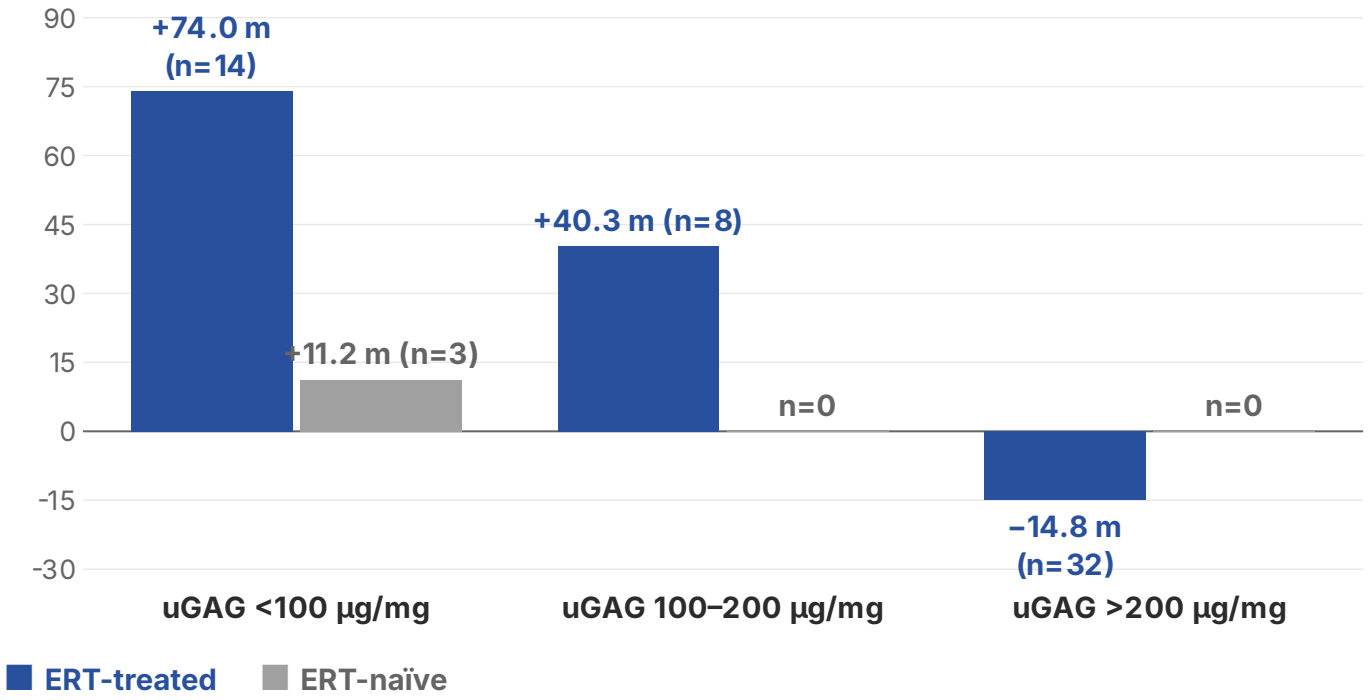
- Under 13: FVC +68% (P=0.0002); FEV1 +55% (P=0.0005)
- 13 and over: FVC +13% (P=0.011); FEV1 stable[1]

a. In the phase 3 trial, FVC and FEV1 improved significantly by 17% (P=0.009) and 11% (P=0.014) at 96 weeks.[2] Chart adapted from Giugliani et al. 2014. FEV1, forced expiratory volume in 1 second; FVC, forced vital capacity

1. Giugliani R et al. Am J Med Genet A 2014;164A:1953–64. 2. Harmatz P et al. J Inherit Metab Dis 2010;33:51–60.

Galsulfase: walking by baseline uGAG level

Change in distance walked from baseline by baseline uGAG level, metres (mean)



uGAG <100 µg/mg

Significant 19% improvement over baseline (P=0.0033)[1]

uGAG 100–200 µg/mg

11% improvement over baseline*[1]

uGAG ≥200 µg/mg

Reduction in distance walked; no ERT-naïve data to compare[1]

**No P value is available. Chart adapted from Giugliani et al. 2014. ERT, enzyme replacement therapy; uGAG, urinary glycosaminoglycans*

1. Giugliani R et al. Am J Med Genet A 2014;164A:1953–64.

Galsulfase: safety data from phase 3 clinical trials

56%

33 of 59 patients had an infusion-associated reaction[1]

98%

53 of 54 patients developed IgG antibodies[1]

Week 1–146

when first infusion reactions appeared[1]

Week 1 — Week 146

Most common adverse reactions[1]

- Pyrexia
- Rash
- Pruritus
- Urticaria
- Chills or rigors
- Nausea
- Headache
- Abdominal pain
- Vomiting
- Dyspnoea

Serious adverse reactions[1]

- Laryngeal oedema
- Apnoea
- Pyrexia
- Urticaria
- Respiratory distress
- Angioedema
- Asthma
- Anaphylactoid reaction

Symptoms of infusion reactions[1]

Pruritus, vomiting, abdominal pain, nausea, hypertension, headache, chest pain, erythema, cough, hypotension, angioedema, respiratory distress, tremor, conjunctivitis, malaise, bronchospasm, arthralgia

IgG, immunoglobulin G

1. Galsulfase Summary of Product Characteristics.